
How Much Can LLMs Hallucinate? An Upper Bound on Goal-Coupling Displacement

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Abstract

1 Hallucination theory bounds *frequency* from below — calibrated LLMs must hal-
2 lucinate at some rate. We bound from above a complementary quantity: the goal-
3 coupling-induced displacement of the post-update belief state — the operational
4 shape of *sycophancy*^[1], where LLM responses match user beliefs over truthful ones.
5 The chain rule of relative entropy on the post-update law equals the *transferred*
6 goal-information that enters the model state, delivering

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \sqrt{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}.$$

7 Čencov’s 1982 uniqueness theorem under categorical parameterization-invariance
8 pins the Fisher-Rao geometry as the canonical metric assignment, with two universal
9 constants on the resulting bound: $C = 2$ globally on the ambient spherical-arc pseu-
10 dometric — the operating constant for adversarial / rare-high-KL prompts including
11 jailbreaks and persona injection (via Rényi-1/2 monotonicity + chord-arc on the unit
12 L^2 -sphere; sharp via a symmetric N -point witness) — tightening to $C = \sqrt{2}$ locally
13 under (R) + (K) + uniform-locality on goal-conditional slices (first-moment sharp
14 via a symmetric two-point witness). A no-go theorem makes the parameterization-
15 invariance commitment load-bearing. A parallel transport-inequality track recovers
16 the canonical Stuart-school cascade form $\propto L_{\text{post}}^2 / \rho_{\text{noise}}$ across the Strand 2 hypothe-
17 sis space (tight on conjugate-Gaussian instances). Operationally the bound factors
18 as coupling strength $\times$ residual ambiguity, with architectures partitioning into a Sep-
19 arated / Partial / Coupled ladder; transformer attention and modern autoregressive
20 sequence models (linear attention, Mamba, RWKV, RetNet, long-convolutions)
21 are structurally Coupled. Binary-uniform two-goal probing gives a closed-form
22 Jensen-Shannon-divergence estimator of the transferred information. The result is
23 a conditional theory of goal-induced displacement on belief-state representations
24 under named regularity hypotheses, not a semantic metric on false outputs.

1 Introduction

26 Recent hallucination theory bounds the *frequency* with which calibrated language models must
27 produce false outputs as a consequence of training-set structure or calibration constraints —
28 Kalai and Vempala 2024^[2] on singleton-fact rates, Kalai et al. 2026^[3] on training-and-evaluation-
29 incentive origins, Karbasi et al. 2025^[4] on detection impossibility, Wu et al. 2025^[5] on VC-
30 dimension obstructions, Suzuki et al. 2025^[6] on probabilistic negligibility. Hallucination in this
31 lineage is treated as architecture-agnostic.

32 Bayesian inverse-problems theory in the lineage of Stuart 2010^[7] has spent fifteen years build-
33 ing Lipschitz-stability bounds on the posterior under perturbations of the prior, the likeli-
34 hood, and the data — Sprungk 2020^[8], Cvetković and Lie 2025^[9], Dolera and Mainini 2023^[10],

35 Garbuno-Iñigo et al. 2023^[11], Hosseini et al. 2025^[12]. The transport-inequality apparatus driving
 36 these bounds — Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze — is mature and verified across
 37 decades of inverse-problems practice.

38 These lineages leave open the implicit belief-goal-coupling structure of attention-based architectures
 39 — a structure intrinsic to LLMs by construction (Lemma E.7), not an incidental property of training.

40 Engaging this structure directly reframes the question: how much can an LLM be pulled away from
 41 what its evidence supports by the goal it is asked to satisfy? The empirical referent is *sycophancy*
 42 — LLM responses matching user beliefs over truthful ones, demonstrated across state-of-the-art
 43 assistants^[1] — read as a goal-coupling-induced displacement of the response distribution from its
 44 evidence-marginal baseline; this paper bounds the magnitude of that displacement.

45 “*Hallucination size*,” in this paper, means the goal-coupling-induced displacement of the model’s
 46 evidence-conditioned belief state — not the semantic distance from a false output to truth. The
 47 two are linked through the in-context-learning correspondence (with the scope and caveats noted
 48 at Appendix E.6 and Section 6) but are not identical objects. The frequency lineage and the size
 49 axis composed here are *complementary*: a coupled-architecture model in (PI)-compatible operating
 50 regime can simultaneously have an inevitable-frequency hallucination floor *and* a bounded-magnitude
 51 goal-coupling displacement on each event.

52 **This paper composes the two** Apply the chain rule of relative entropy directly to the *post-update*
 53 law, marginalized over the goal: $\mathbb{E}_G [\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \| P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+}|e_{\tau}, M_{\tau-})$.
 54 The right-hand side is the *transferred* goal-information that actually enters the post-update model
 55 state. Combined with a Fisher-Rao second-order expansion on the post-update statistical manifold,
 56 this delivers an upper bound on goal-conditional displacement:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+}|e_{\tau}, M_{\tau-})}. \quad (1)$$

57 Under a parameterization-invariance commitment at full Markov-morphism strength, Čencov’s 1982
 58 uniqueness theorem forces Fisher-Rao geometry with constant $C = 2$ globally on the ambient
 59 spherical-arc pseudometric — the operating constant for adversarial / rare-high-KL prompts (sharp via
 60 a symmetric N -point witness) — tightening to $C = \sqrt{2}$ locally under additional Riemannian structure
 61 + KL-coordinate matching + uniform-locality on goal-conditional slices. *Universal, dimension-free,*
 62 *no domain-specific parameters*. A scale-family no-go theorem rules out any coordinate-independent
 63 universal constant for Euclidean chart norms — making the (PI) commitment load-bearing, not
 64 coincidental. We call this *Track 2*.

65 A parallel *Track 1* derivation via a Talagrand transport inequality on the same post-update law
 66 generalizes the canonical Stuart-school cascade. Under Stuart-school Lipschitz-posterior hypotheses
 67 on the deterministic update map plus standard concentration on the goal-induced parameter measure
 68 and slice-wise sub-Gaussianity of the conditional data law, a Lipschitz pushforward + tensorization of
 69 T_2 (Appendix E.2) gives a CT_2 of the canonical Stuart-school form $\propto L_{\text{post}}^2 / \rho_{\text{noise}}$ on the post-update
 70 model law, with $C = \sqrt{CT_2}$ on W_2 — tight at $2L_{\text{post}}^2 / \rho_{\text{noise}}$ on conjugate-Gaussian and exponential-
 71 family-conjugate instances, with an absolute multiplicative overhead from the kernel-tensorization
 72 step in general. The Class 1 (Separated) specialization of Track 1’s hypothesis space is the Stuart-
 73 school setup; Track 1 extends the cascade to the goal-coupled regime outside Stuart-school’s framing.
 74 Track 2’s universal constant is what makes the composition novel; Track 1 connects directly to the
 75 Bayesian-inverse-problems lineage at the cascade level. Both flow from the same chain-rule move on
 76 the post-update law.

77 **The architectural reading** Under a coupling-attenuation hypothesis (H_{κ}), the bound factors:

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} | e_{\tau}, M_{\tau-})}, \quad (2)$$

78 into an *architectural coupling factor* $\kappa_{\text{processing}}$ (how much of the available goal-information the
 79 architecture’s update mechanism transfers) and a *residual ambiguity factor* $I(G; \Omega | e, M)$ (how
 80 much the goal could in principle resolve about the latent world-state beyond what the evidence already
 81 pins down). Architectures partition by their goal-update topology into a monotonic ladder — Class

1 (Separated), Class 2 (Partial), Class 3 (Coupled). For Class 1, $\kappa_{\text{processing}} \leq 1$ holds automatically by data-processing on the Markov chain $G \rightarrow \Omega \rightarrow M_{\tau+}$. For Class 2 and Class 3, (H_{κ}) is a named structural commitment with documented sufficient conditions and a worst-case witness (parrot architecture, $\kappa_{\text{processing}} = \infty$) excluded by scope.

Plain decoder-only transformer attention is *Class 3 (Coupled) by construction* in a precise structural sense: every output computation downstream of the goal positions has a directed-graph path from at least one goal position whenever attention weights are non-degenerate on the goal row (Lemma E.7, by induction on layer depth, robust to RMSNorm / FlashAttention / causal masking / sliding-window). The structural classification is at the *architectural* level — what attention *can* do; the *quantitative* coupling magnitude $\kappa_{\text{processing}}$ requires additional assumptions and is empirically estimated, not analytically derived. The structural Coupled-class claim does not depend on the token-distribution-as-belief-state idealization that the bias-bound *application* additionally invokes via the in-context-learning correspondence^[13–16].

Contributions

1. **An architectural classification.** Goal/Update Coupling Class — Class 1 (Separated), Class 2 (Partial), Class 3 (Coupled) — partitions architectures monotonically by goal-update topology, grounded in the Pearl-blanket reading^[17] of the Markov-blanket apparatus. Lemma E.7 derives Class 3 status for transformer attention from connectivity alone.
2. **An umbrella upper bound** (1) on goal-conditional displacement of the post-update model state, derived via chain rule on the post-update law. Two named tracks supply C — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2).
3. **A no-go that forces a coordinate commitment.** No coordinate-independent universal constant exists for Euclidean chart norms (a scale-family construction). Under (PI) alone, Rényi-1/2 monotonicity + the chord-arc identity give a sharp universal $C = 2$ globally on the ambient spherical-arc pseudometric; under the additional (R) + (K) triple plus uniform-locality, Čencov’s uniqueness theorem tightens this to Fisher-Rao + $\sqrt{2}$ locally — both forced, neither free.
4. **An architectural-corollary form** (2) under (H_{κ}) , recovering the $\kappa \times \mathcal{A}$ factorization that gives the bound its operational reading.

Section 3 sets up the architectures and the (PI), (R), (K) axioms. Section 4 states the umbrella theorem with both tracks, the no-go, and the architectural corollary. Section 5 sketches the mechanism — both proofs flow from the post-update chain rule. Limitations and the bound’s relation to the frequency-lower-bound lineage are in Section 6.

2 Related Work

Two literatures sit on either side of the gap this paper bridges; brief categorization (full positioning across all six adjacent strands in Appendix F).

LLM hallucination theory has produced architecturally-agnostic frequency lower bounds on calibrated language models — calibration-constrained singleton-fact rates, training-and-evaluation-incentive origins, learnability and computability obstructions, probabilistic-negligibility positive results (cited in §1). These results derive from training-set structure or capacity considerations and do not engage the architecture’s belief-goal-coupling structure. They bound *how often* a calibrated model must hallucinate.

Bayesian inverse-problems posterior stability (the Stuart school) has produced mature Lipschitz-continuity bounds on the posterior under perturbations of the prior, the likelihood, and the data, using transport-inequality machinery (Otto-Villani, Bakry-Émery, Gozlan, Bobkov-Götze) and Fisher-information-flavored constants in optimal-transport frameworks (cited in §1). These results address well-posedness of Bayesian inference; they have not been applied to belief-goal-coupled inference. The machinery bounds *how stable* a Bayesian posterior is to specified perturbations.

The two bound different things, sit in different research traditions, and do not engage with each other. The chain-rule of relative entropy applied directly to the post-update law (Lemma 5.1) is the bridging move: it converts the goal-coupling problem into a transferred-information quantity, which

the Stuart-school transport machinery and the Fisher-Rao expansion both bound at the post-update law. *Both* lineages enter — Track 1 inherits the Otto-Villani / Bakry-Émery cascade; Track 2 deploys Čencov uniqueness on the manifold the Stuart school operates on. The composition is the contribution; the architectural reading via Class 1 / Class 2 / Class 3 makes the bound operational.

Adjacent strands — Fisher-Rao geometry for Bayesian sensitivity analysis, Owhadi-Scovel-Sullivan brittleness no-gos, mean-field variational inference stability, the Markov-blanket apparatus from active inference — sit close enough to require explicit cite-and-distinguish but do not resolve the gap. Detailed positioning across all strands is in Appendix F.

3 Setup

We work with goal-conditioned probabilistic models in a Bayesian-update setting. The state at time τ^- before update consists of an epistemic component M_{τ^-} (the model state, an element of a parameter manifold $\mathcal{M}$) and a goal component G . An event e_τ arrives — a token, an observation, a query response — and the model updates to $M_{\tau+}$. The latent world-state Ω_τ is what the event is evidence about. The architecture’s update mechanism produces a post-update model state $M_{\tau+}$ whose law depends — possibly stochastically — on the goal G . The goal-conditional post-update law is $P_{M_{\tau+}|e_\tau, M_{\tau^-}, G}$; the goal-marginal post-update law $P_{M_{\tau+}|e_\tau, M_{\tau^-}}$ is the natural baseline against which goal-coupling is measured (the operating-distribution average over goals).

3.1 The bias quantity

The *goal-conditional bias* measures the displacement of the goal-conditional law from the goal-marginal:

$$\|\Delta M_{\text{bias}}(G)\| := d_{\mathcal{M}}\left(P_{M_{\tau+}|e_\tau, M_{\tau^-}, G}, P_{M_{\tau+}|e_\tau, M_{\tau^-}}\right), \quad (3)$$

a random variable in G . Throughout, $d_{\mathcal{M}}(\cdot, \cdot)$ denotes one of two distinct objects, used cleanly within their respective tracks. (*Track 1*) the Wasserstein distance W_2 on **laws over the parameter manifold** $\mathcal{M}$ — the goal-conditional and goal-marginal post-update random elements of $\mathcal{M}$ are compared as probability laws on $\mathcal{M}$ with a ground metric on $\mathcal{M}$. (*Track 2*) the Fisher-Rao spherical-arc distance d_{FR} on the statistical-manifold sub-case (H1), where each model state $M_t \in \mathcal{M}$ carries an associated probability distribution P_{M_t} over the latent world-state — and the bias quantity is the random spherical arc between $P_{M_{\tau+}|e_\tau, M_{\tau^-}, G}$ and $P_{M_{\tau+}|e_\tau, M_{\tau^-}}$ as points on the unit L^2 -sphere of $\sqrt{p}$. We adopt the *Amari-Nagaoka spherical-arc convention* $d_{FR}(P, Q) := 2 \arccos \int \sqrt{pq}$ throughout, with $d_{FR} \in [0, \pi]$. (Equal to the intrinsic Fisher-Rao geodesic on the full simplex / nonparametric ambient space; on a parametric submanifold the ambient spherical arc is the chart-invariant pseudometric we bound.) The locally-tight $\sqrt{2}$ regime (under (PI)+(R)+(K)+(H4')) operates in the small-displacement limit where the ambient arc agrees with submanifold-intrinsic geodesics at second order; the globally-valid $C = 2$ regime (under (PI) alone) operates on the ambient arc directly. Euclidean-on-parameters is the choice Section 4.1 rules out.

3.2 Goal/Update Coupling Class

Architectures partition by whether G has a causal path into the belief-update computation. The partition is monotonic in coupling strength.

Class 1 (Separated) is the architectural-separation case studied throughout Bayesian inverse-problems theory. Class 3 (Coupled) is the case the existing hallucination-theory lineage doesn’t engage geometrically. Class 2 (Partial) is genuinely intermediate — hybrid LLM systems with retrieval, tools, scratchpads, or external memory — and lives at the system level rather than the architecture level. The Pearl-blanket reading^[17] of the Markov-blanket apparatus grounds the partition; we do not adopt the Friston-blanket metaphysical reading.

The bound’s architectural-corollary form (2) factors transferred goal-information through an *attenuation ratio* — the proportion of in-principle-resolvable goal-information that the architecture’s update mechanism actually transfers into the post-update model state:

Table 1: Goal/Update Coupling Class — partition by conditional-independence structure.

Class	Goal/update coupling	Examples
Class 1 (Separated)	Holds by construction — estimator has no causal path from G	Kalman filter + LQR; modular RL with separate world model
Class 2 (Partial)	Holds for some pathways, fails for others	Biological cortex; hybrid systems with separate preprocessing
Class 3 (Coupled)	Fails by construction — G is causally upstream of every downstream output computation	Autoregressive sequence models with causally-aggregated cross-position mixer (transformers, linear attention, Mamba/SSMs, RWKV, RetNet, long-convolutions — Lemma E.7 + Corollary E.8)

$$\kappa_{\text{processing}} := \frac{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}{I(G; \Omega_{\tau} \mid e_{\tau}, M_{\tau-})}. \quad (4)$$

179 Conditioning on $M_{\tau-}$ matters: prior correlation between goals and prior model state — present even
 180 in Class 1 — would inflate the measure otherwise. *Class 1 formally:* conditional on $(e_{\tau}, M_{\tau-})$, $G \perp\!\!\!\perp$
 181 $M_{\tau+} \mid \Omega_{\tau}$ — G affects $M_{\tau+}$ only through the latent world-state Ω_{τ} . This conditional independence
 182 is the formal content of “estimator has no causal path from G ” and gives the Markov chain $G \rightarrow$
 183 $\Omega_{\tau} \rightarrow M_{\tau+}$ (conditional on $(e_{\tau}, M_{\tau-})$), to which the data-processing inequality applies, yielding
 184 $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 and Class 3, $\kappa_{\text{processing}} \leq 1$ is a named structural commitment;
 185 the parrot architecture $M_{\tau+} := G$ with $G \perp\!\!\!\perp \Omega_{\tau}$ given $(e_{\tau}, M_{\tau-})$ produces $\kappa_{\text{processing}} = \infty$, explicitly
 186 outside scope. *Convention.* When both numerator and denominator vanish (no transferred goal-
 187 information *and* no residual ambiguity, e.g., a Class 1 architecture on goal-orthogonal evidence), set
 188 $\kappa_{\text{processing}} := 0$; the bound’s right-hand side is then trivially zero.

189 3.3 Axioms

190 The universal-constant route (Section 5.2, Track 2) commits to three axioms on the bound’s metric
 191 $d_{\mathcal{M}}$:

192 **Hypothesis 3.1** ((PI) Sufficient-statistic invariance across the natural category). *The bias bound’s*
 193 *distance / divergence is a natural assignment on the category of standard parametric statistical*
 194 *models — for each model, a candidate distance or divergence — commuting with congruent Markov*
 195 *morphisms (sufficient statistics) in the sense of Čencov 1982^[18]. The induced object on $\mathcal{M}$ inherits*
 196 *this categorical assignment.*

197 (PI) at full Markov-morphism strength is the *categorical* invariance Čencov’s uniqueness theorem
 198 requires: the assignment must respect every sufficient-statistic morphism across the category, not
 199 merely smooth coordinate changes on a single fixed $\mathcal{M}$. (PI) alone admits both Riemannian metrics
 200 and f -divergences as candidates (the global Fisher-Rao spherical-arc bound and the Hellinger backstop
 201 in Appendix D are the two (PI)-only theorems); (R) below adds the Riemannian-metric specifier
 202 needed for Čencov’s local uniqueness statement.

203 **Hypothesis 3.2** ((R) Riemannian structure). *The bound’s metric $d_{\mathcal{M}}$ derives from a smooth Riemannian*
 204 *metric on $\mathcal{M}$.*

205 (R) restricts the (PI)-invariant candidates to Riemannian metrics specifically, filtering out divergence-
 206 form alternatives (TV, χ^2 , Hellinger-as-divergence) that are (PI)-invariant as f -divergences but are
 207 not metric distances in the Riemannian sense.

208 **Hypothesis 3.3** ((K) KL second-order matching). $\text{KL}(P_{\theta} \parallel P_{\theta+\delta}) = \frac{1}{2} d_{\mathcal{M}}^2(P_{\theta}, P_{\theta+\delta}) + O(d_{\mathcal{M}}^3)$.

209 (K) fixes the global scalar Čencov leaves free, pinning the bound’s constant C at $\sqrt{2}$. (R) and (K)
 210 are both implicit in any “ $\mathbb{E} d \leq C\sqrt{T}$ ” theorem-shape with second-order matching; we name them
 211 explicitly so the uniqueness statement of Theorem 4.5 has its hypotheses fully stated.

The transport-inequality route (Section 5.1, Track 1) and the dimension-restricted slice-wise refinements (Theorem 4.1’s Track 2 locally-tight instantiation) introduce additional regularity hypotheses inline with the theorems that invoke them. Throughout the paper we restrict to the *statistical-manifold sub-case*:

Hypothesis 3.4 ((H1) Statistical-manifold sub-case). *The model state $M_t \in \mathcal{M}$ corresponds to a probability distribution P_{M_t} over latent world-states, and $\mathcal{M}$ is locally a statistical manifold in the sense of Amari and Nagaoka 2000^[19]; $\mathcal{M}$, the goal-space, and the event-space are standard Borel, so regular conditional probabilities exist as Markov kernels^[20] Theorem 6.3.*

(H1) is satisfied by parametric Bayesian posterior families, exponential families, conjugate priors, and the parametric-belief-state subclass of LLMs that admits a natural distributional reading. The standard-Borel regularity is what makes the post-update chain rule (used in both Tracks) hold in the abstract-spaces form^[21] Theorem 3.4; ^[22] Theorem 5.4.

Coupled-class autoregressive connectivity (deferred). Lemma E.7 establishes that plain decoder-only transformer attention — and the broader autoregressive-sequence-model family with causally-aggregated cross-position mixers under a per-source non-degeneracy condition (linear attention, Mamba/SSMs, RWKV, RetNet, long-convolutions; named-architecture verifications + ICL-correspondence caveat in Appendix E.6) — is structurally Class 3 by directed-graph reachability. The claim is on $\kappa_{\text{processing}}$ ’s *support*, not its magnitude.

4 Main Results

The umbrella theorem bounds the goal-conditional displacement of the post-update model state by transferred goal-information. Two named tracks supply the constant — a transport-inequality cascade (Track 1) and a Fisher-Rao geometry route (Track 2) — both flowing from a common chain-rule move on the post-update law (mechanism in Section 5).

Theorem 4.1 (Umbrella bound on goal-conditional displacement). *Under (H1) and the regularity hypotheses below, for fixed event-prior pair $(e_\tau, M_{\tau-})$ with $\mathbb{E}$ taken over $G \sim P(G | e_\tau, M_{\tau-})$:*

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \cdot \sqrt{I(G; M_{\tau+} | e_\tau, M_{\tau-})}, \quad (5)$$

with C supplied by either of two routes:

(Track 1, transport-inequality): *Under additional (H2') — a Talagrand T_2 inequality on the post-update law — $C = \sqrt{C_{T_2}}$ with metric W_2 . The constant C_{T_2} recovers the canonical Stuart-school cascade value $2L_{\text{post}}^2 / \rho_{\text{LSI}}$ as a special case under standard log-Sobolev concentration.*

(Track 2, Fisher-Rao spherical-arc): *The bound holds with metric d_{FR} in the ambient Amari-Nagaoka spherical-arc convention of Section 3.1 ($d_{FR}(P, Q) = 2 \arccos \int \sqrt{pq}$, the great-circle arc on the unit L^2 -sphere of $\sqrt{p}$). Two universal forms — both dimension-free, neither carrying domain-specific parameters:*

locally tight — *under (PI)+(R)+(K) on the natural categorical metric assignment plus (H4') at parameter δ_* , $C = \sqrt{2(1 + R(\delta_*))}$ with explicit $R(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$ (locally tight at $\sqrt{2}$ in the limit; Čencov uniqueness pins the metric and constant via Theorem 4.5);*

globally valid — *under (PI) alone (no (R), no (K), no (H4')), the ambient spherical-arc bound $C = 2$ holds throughout — sharp via a symmetric N -point witness as $N \rightarrow \infty$ (Theorem D.1 in Appendix D; the $\sqrt{2}$ overhead is exactly the chord-arc factor at the unit L^2 -sphere’s antipode in squared-distance form). On parametric submanifolds the ambient spherical arc is the chart-invariant pseudometric; submanifold-intrinsic geodesics may be longer (the local-tight regime is exactly where ambient and intrinsic agree at second order).*

Hypothesis 4.2 ((H2') Talagrand T_2 on the post-update law). $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})$ for all P . *Sufficient conditions in Appendix B (LSI, Otto-Villani; dimension-free sub-Gaussian Lipschitz concentration^[23]).*

Hypothesis 4.3 ((H4') Uniform local regime). $\text{ess sup}_g \text{KL}(P_{M_{\tau+}|G=g} \| P_{M_{\tau+}}) \leq \delta_*$ for some $\delta_* \in (0, 1)$ — *every goal-conditional slice uniformly close to the goal-marginal, not merely the*

average (strictly stronger than small- I_M). Adversarial / rare-high-KL prompts (jailbreaks, persona injection) exit ($H4'$); the global $C = 2$ bound (Theorem D.1) is the operating tool there.

261 **Remarks on Theorem 4.1.**

262 *The two tracks bound different metrics with the same square-root-in-information shape.* Track 1 lives on
263 W_2 , Track 2 on d_{FR} . Both deliver bounds of form $\mathbb{E} d \leq C\sqrt{I_M}$ where $I_M = I(G; M_{\tau+} | e_{\tau}, M_{\tau-})$
264 is transferred goal-information. The two are not “different scalings”; they are bounds on different
265 metrics with the same information dependence.

266 *The bound is unconditional in the architectural classification.* Both Tracks bound transferred goal-
267 information directly. (H1)–(H2') and (H1)–(H4')–(PI)–(R)–(K) are regularity hypotheses on the
268 post-update law and the metric. Neither makes a structural commitment about how the goal G enters
269 the architecture’s update mechanism. The architectural reading enters at corollary level via (H_{κ})
270 below.

271 *Track 2’s constant is universal and dimension-free.* Track 1’s C_{T_2} is family-specific (depends on
272 the post-update concentration). Track 2’s $\sqrt{2}$ is the same constant across the entire categorical
273 metric assignment under the (PI)+(R)+(K) triple, by Čencov uniqueness (Theorem 4.5 below); on the
274 parametric submanifold of any specific application, the induced metric is Fisher-Rao with the same
275 constant.

276 *Track 1 generalizes the Stuart-school cascade.* Under Stuart-school Lipschitz-posterior hypothe-
277 ses^[7,8,10,12] plus T_2 concentration on the goal-induced parameter measure and slice-wise conditional
278 sub-Gaussianity, Theorem E.3 yields the canonical $C_{T_2} \propto L_{\text{post}}^2 / \rho_{\text{noise}}$ — tight at $2L_{\text{post}}^2 / \rho_{\text{noise}}$ on
279 conjugate-Gaussian and exponential-family-conjugate instances (Appendix C); Track 1 extends the
280 cascade beyond Class 1 (Separated) to goal-coupled architectures, a regime Stuart-school does not
281 address.

282 **4.1 A no-go that forces (PI)**

283 The universal-constant route (Track 2’s $\sqrt{2}$) is not coincidental. A scale-family construction shows
284 that no coordinate-independent universal constant exists for Euclidean chart norms — making the
285 (PI) commitment load-bearing.

286 **Theorem 4.4** (No-go on Euclidean chart norms). *Suppose there were a constant $C_0 < \infty$ —*
287 *independent of any chart on any statistical manifold — such that for every chart ϕ on $\mathcal{M}$ and*
288 *every goal-coupled architecture satisfying (H1)+(H2') with non-zero post-update displacement,*
289 $\mathbb{E} \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}$. *Then no such C_0 exists.*

290 The proof is short: a chart rescaling $\phi \mapsto a\phi$ scales the chart-Euclidean Wasserstein distance linearly
291 while leaving the right-hand side (chart-invariant information) unchanged; taking $a \rightarrow \infty$ contradicts
292 the fixed $C_0\sqrt{I}$. Full statement and proof in Appendix E.4. The Gaussian scale family with chart σ
293 versus $\log \sigma$ is the cleanest illustration.

294 **Remark** Theorem 4.4 is *adjacent in shape but opposite in direction* to Owhadi-Scovel-Sullivan’s
295 Bayesian-brittleness results^[24,25]. Their no-go is for *arbitrary stability under finite information* in
296 fixed metrics; ours is for *a universal constant absent a coordinate-invariance commitment*, in a setting
297 where (H2')’s post-update concentration explicitly excludes their brittleness regime. The two no-gos
298 constrain different things; neither implies the other.

299 **4.2 Čencov uniqueness and sharpness**

300 The no-go requires a coordinate-invariance commitment if a universal constant is wanted at all;
301 (PI)+(R)+(K) is the natural commitment for an information-coordinate cascade; and Čencov unique-
302 ness within that commitment leaves little further freedom.

303 **Theorem 4.5** (Uniqueness and sharpness of the Fisher-Rao $+\sqrt{2}$ bound (local)). *Under*
304 *(H1)+(PI)+(R)+(K)+(H4'), in the umbrella form (5) (where $\|\Delta M_{\text{bias}}\|$ is the $d_{\mathcal{M}}$ -displacement*
305 *per (3)): (a) $d_{\mathcal{M}}$ is uniquely the ambient Amari-Nagaoka Fisher-Rao spherical-arc distance, agreeing*
306 *with submanifold-intrinsic Fisher-Rao geodesics at second order in the ($H4'$) regime; (b) $C = \sqrt{2}$ is*
307 *the unique sharp upper-bound constant — no $C < \sqrt{2}$ holds uniformly across the ($H4'$) regime.*

Part (a) follows from Čencov’s 1982 uniqueness theorem applied to the natural categorical metric assignment of (PI). Part (b) is verified by a symmetric two-point witness on the conjugate-Gaussian Class 1 family: $G \in \{-a, +a\}$ uniform delivers $\mathbb{E} d_{FR}/\sqrt{I_M} \rightarrow \sqrt{2}$ as $a/\sigma \rightarrow 0$ (the symmetry forces d_{FR} exactly constant across slices, so Jensen is tight). Full proof + witness construction in Appendix E.5. Alternative chart-independent commitments (TV, Hellinger-as-divergence, χ^2 , Rényi) are all defeated by (R)/(K) or non-tightness; the no-go narrows the candidate space, and (PI)+(R)+(K) selects Fisher-Rao + $\sqrt{2}$.

4.3 The architectural corollary

The umbrella theorem bounds displacement by *transferred* goal-information. The architectural reading factors this transferred information into a structural-architectural quantity (where can the goal flow?) and an information-theoretic quantity (how much could the goal in principle resolve?).

Hypothesis 4.6 ((H_κ) Bounded architectural attenuation). $\kappa_{\text{processing}} \leq \kappa^* < \infty$, with κ^* a named architectural constant (typically 1 for non-amplifying architectures).

Corollary 4.7 (Architectural factorization). Under (H1) + the Track-specific hypotheses + (H_κ) at level κ^* , the bound factors as

$$\mathbb{E} \|\Delta M_{\text{bias}}\| \leq C \sqrt{\kappa^* \cdot I(G; \Omega_\tau \mid e_\tau, M_{\tau^-})}, \quad (6)$$

with C supplied by Track 1 or Track 2 as in Theorem 4.1.

Remarks on Corollary 4.7.

The factorization is the contribution-headline. Coupling strength $\kappa_{\text{processing}} \times$ residual ambiguity $I(G; \Omega \mid e, M)$ — architecture $\times$ information. Theorem 4.1’s unconditional form is the structural backbone whose math holds without (H_κ) ; the corollary is the operational reading.

Class 1 (Separated) gives (H_κ) automatically. The Markov chain $G \rightarrow \Omega_\tau \rightarrow M_{\tau+}$ (conditional on $(e_\tau, M_{\tau-})$) plus data-processing inequality yields $\kappa_{\text{processing}} \leq 1$ by construction. (H_κ) is a derived consequence, not a commitment, for Class 1.

Class 2 (Partial) and Class 3 (Coupled) require (H_κ) as a named structural commitment. Two clean sufficient conditions and a parrot-architecture worst-case witness ($\kappa_{\text{processing}} = \infty$ when $M_{\tau+} := G$ on goals orthogonal to the latent world-state) are documented in Appendix B.3. The unconditional theorem still bounds parrot-architecture displacement directly by transferred information; the $\kappa \times \mathcal{A}$ factorization just doesn’t apply.

Transferred information has a closed-form empirical estimator. For $G \in \{g_1, g_2\}$ uniform, $I(G; M_{\tau+} \mid e_\tau, M_{\tau-}) = \text{JSD}(P_{M_{\tau+}|g_1}, P_{M_{\tau+}|g_2})$ — Monte-Carlo estimable from per-goal response samples; the local Track-2 form gives $\mathbb{E} \|\Delta M_{\text{bias}}\| \leq \sqrt{2 \text{JSD}}$, the global form $\mathbb{E} d_{FR} \leq 2\sqrt{\text{JSD}}$ (Appendix D). Factorization additionally requires lower-bounding $I(G; \Omega \mid e, M_{\tau-})$ — architecture-side modeling, Section 6.

5 Mechanism

Both tracks bridge to the bound through the same move: a chain rule of relative entropy applied directly to the post-update law, marginalized over the goal. This is the structural composition of the two literatures — the Bayesian-inverse-problems posterior-stability machinery and the architectural classification meet at this identity.

Lemma 5.1 (Chain rule on the post-update law). Under (H1), $\mathbb{E}_G [\text{KL}(P_{M_{\tau+}|e, M_{\tau-}, G} \parallel P_{M_{\tau+}|e, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-})$.

The identity is exact (Polyanskiy and Wu²¹, Theorem 3.4; Gray²², Theorem 5.4 for the abstract-spaces form). The right-hand side is *transferred* goal-information — the actually-realized coupling between G and the post-update model state. Combined with either a transport inequality on the post-update law or a Fisher-Rao second-order expansion of KL, the bound follows by taking expectation over G and applying Jensen.

353 5.1 Track 1 — transport-inequality cascade

354 Apply (H2') slice-wise: $W_2^2(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \leq C_{T_2} \cdot \text{KL}_g$. Take expectation over G , substitute
 355 Lemma 5.1 for the slice-wise mean, and apply Jensen on $\sqrt{\cdot}$: $\mathbb{E} W_2 \leq \sqrt{C_{T_2} I_M}$. The cascade
 356 composes textbook information-theoretic and transport-inequality machinery — Otto-Villani for the
 357 LSI-implies- T_2 direction, Bakry-Émery for K -strong-log-concavity-implies-LSI, Gozlan’s dimension-
 358 free sub-Gaussian Lipschitz-concentration characterization. The contribution is the application to the
 359 goal-conditional bias quantity on the post-update law: there is no separate posterior-pushforward step,
 360 because the chain rule lives on the post-update law directly. Full proof + (H2') sufficient-condition
 361 verifications in Appendix B.

362 5.2 Track 2 — Fisher-Rao expansion

363 The KL-to-Fisher-Rao second-order expansion on the ambient L^2 -sphere of $\sqrt{p}$ — $\text{KL}(P \parallel Q) =$
 364 $\frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$ in the Amari-Nagaoka spherical-arc convention (Amari and Nagaoka¹⁹ §3.7)
 365 — applies to nearby points on the ambient sphere. The ambient framing absorbs the goal-marginal-
 366 baseline-as-mixture concern: both the slice law $P_{M_{\tau+}|G=g}$ and the goal-marginal mixture $P_{M_{\tau+}}$ are
 367 points on the ambient sphere by construction. Apply slice-wise under (H4'), take $\mathbb{E}_G$, substitute
 368 Lemma 5.1, and apply Jensen on $\sqrt{\cdot}$: $\mathbb{E} d_{FR} \leq \sqrt{2} \sqrt{I_M} (1 + o(1))$. The $(1 + o(1))$ remainder is
 369 controlled uniformly under (H4') by the Amari-Chentsov tensor and minimum Fisher eigenvalue (full
 370 bound in Appendix E.3). The constant $\sqrt{2}$ is what (PI)+(R)+(K) buys: Čencov uniqueness pins it
 371 (Theorem 4.5).

372 *Outside (H4')*. Track 2 has globally-valid companion bounds (ambient FR spherical-arc at $C = 2$,
 373 Hellinger chord at $1/\sqrt{2}$) under (H1)+(PI) only — full taxonomy and chord-arc-at-antipode geometric
 374 reading in Appendix D.

375 5.3 No-go — chart rescaling

376 The no-go’s proof is a chart-rescaling argument: $\phi \mapsto a\phi$ scales the chart-Euclidean W_2 linearly while
 377 leaving KL, MI, Fisher-Rao spherical-arc, and Hellinger chart-invariant; taking $a \rightarrow \infty$ contradicts
 378 any candidate fixed $C_0\sqrt{I}$. Gaussian scale charts σ vs. $\log \sigma$ illustrate. Constructively: universal
 379 constants require coordinate-invariant norms — under (PI)+(R)+(K), Čencov uniqueness selects
 380 Fisher-Rao + $\sqrt{2}$ (Appendix E.4).

381 6 Conclusion

382 The opening question — *how much can an LLM be pulled away from its evidence by its goal?* — sits
 383 on an axis the frequency lineage doesn’t engage. A coupled-architecture model can simultaneously
 384 have an inevitable-frequency hallucination floor (training-data structure, calibration, capacity) *and* a
 385 bounded-magnitude goal-coupling displacement on each event — orthogonal axes, complementary
 386 readings.

387 **Limitations** Conditional theorems under named regularity ((H2'), (H4'), (H $_{\kappa}$)) and the (PI) ax-
 388 iomatic commitment, on theory-only grounds — one channel of hallucination, complementary to the
 389 frequency lineage.

390 **Future directions** *First instantiation of $\hat{\kappa}_{processing}$ on a deployed LLM*: the bound becomes op-
 391 erational once κ^* is bounded; binary-uniform two-goal probing has a closed-form JSD estimator
 392 (Theorem 4.1), with activation-level mediation via representation-engineering / steering-vector /
 393 mechanistic-interpretability for the architectural side. *Sycophancy*^[1] — LLM responses matching
 394 user beliefs over truthful ones across state-of-the-art assistants — is the natural empirical referent;
 395 Sharma et al.’s *answer sycophancy metric* is essentially the binary-uniform two-goal probe of Corol-
 396 lary 4.7, so instantiating the bound on existing sycophancy benchmarks would convert them into
 397 magnitude-bound diagnostics. *Bridging to the frequency-lower-bound lineage*: a layered theory com-
 398 posing Kalai-Vempala-style frequency-floors with the goal-coupling magnitude-ceiling here would
 399 give the field a complete two-axis picture.

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A Failed routes

We document two derivation routes that failed at structural levels, so future work does not re-attempt them without new evidence.

A.1 Route F1: Cramér-Rao inversion

Attempt Treat G as a parameter to be estimated from Ω given (e, M) . The Cramér-Rao lower bound gives $\text{Var}(\hat{G}) \geq \mathbf{I}_G^{-1}$ where $\mathbf{I}_G$ is the Fisher information about G given (Ω, e, M) . Invert this to a posterior-sensitivity factor: an upper bound $\|\Delta M_{\text{bias}}\| \leq L_M \cdot \sqrt{\mathbf{I}_G^{-1}}$ via a Lipschitz-in- G constant on the coupled-update mechanism.

Why it fails Cramér-Rao bounds estimator variance from *below*. To produce an *upper bound* on $\|\Delta M_{\text{bias}}\|$ via Cramér-Rao inversion would require fixing an estimator class within which $\mathbf{I}_G^{-1}$ is also an upper bound on variance — but no such estimator class exists for the coupled architecture’s update mechanism. The mechanism is a deterministic function of (G, e, M) , not an estimator of G , and there is no uniform-class assumption available. The direction of the Cramér-Rao bound and the direction of the bias bound are opposite; inverting one to the other requires an additional assumption that is not available.

Lesson Information-theoretic *lower-bound* machinery for estimator performance does not invert to upper bounds on side-information-induced displacement.

A.2 Route F2: Rate-distortion inversion

Attempt Apply Shannon’s rate-distortion inequality $R(D) \geq h(X) - \frac{1}{2} \log(2\pi e D)$ for a Gaussian source. Invert to $D \leq \sigma^2 \cdot 2^{-2(R-h(X))}$ with “bits in” interpreted as $I(G; \Omega|e, M)$, “distortion” interpreted as $\|\Delta M_{\text{bias}}\|^2$.

Why it fails Rate-distortion theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ; it does not describe the maximum *displacement induced* by injecting side-information into an update mechanism. The two problems have different structures: source-coding is about optimal representations; side-information injection is about perturbation propagation. Identifying “bits in” with I and “distortion” with displacement does not produce a sound inequality — the rate-distortion curve is about a different operational quantity.

Lesson Source-coding theorems are about optimal representations of a source, not spatial displacement induced by side-information injection. Upper bounds on displacement require transport-inequality machinery (Pinsker, Otto-Villani, Bakry-Émery, Lipschitz-posterior).

A.3 The pattern

Both failed routes attempt to invert *information-theoretic lower-bound* machinery (Cramér-Rao for estimator error; rate-distortion for source coding) into *upper bounds on displacement*. The structural mismatch is uniform: lower-bound machinery for optimization quantities does not invert to upper bounds on perturbation quantities without additional structure that is not available in coupled-architecture settings. Transport-inequality machinery is on the other side of this divide and is the correct family for upper-bound-on-displacement results.

B Hypothesis verification details

This appendix collects the regularity-condition verifications referenced in Section 5.1 and Section 5.2.

B.1 (H2') Talagrand T_2 inequality on the post-update law — sufficient conditions

(H2') requires $W_2^2(P, P_{M_{\tau+}|e, M_{\tau-}}) \leq C_{T_2} \text{KL}(P \| P_{M_{\tau+}|e, M_{\tau-}})$ for all probability measures P . Three clean sufficient conditions:

Route (a): LSI on the post-update law If $P_{M_{\tau+}|e, M_{\tau-}}$ satisfies a logarithmic Sobolev inequality with constant $\rho_{\text{post}} > 0$,

$$\text{Ent}_{P_{M_{\tau+}|e, M_{\tau-}}}(f^2) \leq \frac{2}{\rho_{\text{post}}} \mathbb{E}_{P_{M_{\tau+}|e, M_{\tau-}}} [\|\nabla f\|^2] \quad \text{for all sufficiently smooth } f,$$

then by^[26] Theorem 1, (H2') holds with $C_{T_2} = 2/\rho_{\text{post}}$. A sufficient condition for LSI, by Bakry-Émery’s curvature-dimension criterion^[27], is K -strong log-concavity of the post-update density: $-\nabla^2 \log p_{M_{\tau+}|e, M_{\tau-}} \succeq K \cdot I$ for some $K > 0$, giving $\rho_{\text{post}} \geq K$.

Hypothesis (S) — own proof of strong log-concavity for Bayesian post-updates When the post-update law is itself a Bayesian posterior with negative log-likelihood $\Phi(\theta; \omega)$ and prior density $\pi(\theta)$, the posterior negative log-density is $U(\theta; \omega) = \Phi(\theta; \omega) - \log \pi(\theta)$. Strong log-concavity of the posterior is a *direct* statement about the Hessian-in- θ of U : $\nabla_{\theta}^2 U(\theta; \omega) \succeq K_{\text{eff}} \cdot I$ uniformly in $\omega \in \Omega$, for some $K_{\text{eff}} > 0$. Two clean sufficient conditions:

- (S-i) *Convex likelihood + strongly log-concave prior.* If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq 0$ uniformly in (θ, ω) (likelihood log-concave in θ) and π is K -strongly log-concave (so $-\nabla_{\theta}^2 \log \pi \succeq K \cdot I$), then $\nabla_{\theta}^2 U \succeq K \cdot I$, giving $K_{\text{eff}} \geq K$.
- (S-ii) *Bounded negative likelihood curvature dominated by prior curvature.* If $\nabla_{\theta}^2 \Phi(\theta; \omega) \succeq -K_0 \cdot I$ uniformly with $K_0 < K$ (likelihood may be locally non-convex but its negative-curvature is bounded), and π is K -strongly log-concave, then $\nabla_{\theta}^2 U \succeq (K - K_0) \cdot I$, giving $K_{\text{eff}} \geq K - K_0 > 0$.

Either condition gives Bakry-Émery $\rho_{\text{post}} \geq K_{\text{eff}}$ and Otto-Villani $C_{T_2} \leq 2/K_{\text{eff}}$. Sprungk 2020^[8] derives sharper local-Lipschitz constants for the posterior across TV / Hellinger / W_2 / KL; Cvetković and Lie 2025^[9] establish matched lower bounds; Dolera and Mainini 2023^[10] give global W_2 -Lipschitz constants in terms of Fisher-information functionals — which can sharpen C_{T_2} beyond the bare $2/K_{\text{eff}}$. Mixed-Hessian smoothness $\|\partial_{\theta\omega}^2 \Phi\| \leq H$ is *not* a substitute: it controls how the posterior mode shifts with ω , but the LSI / strong-convexity claim is about the curvature in θ and must be made directly.

Route (b) (corrected — bounded support gives T_1 , not T_2). Bare bounded support of diameter D on $\mathcal{M}$ is **not** a T_2 -sufficient condition. *Counterexample.* On $\{0, 1\} \subset \mathbb{R}$ (diameter $D = 1$), let $\nu = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\mu_{\epsilon} = (\frac{1}{2} - \epsilon)\delta_0 + (\frac{1}{2} + \epsilon)\delta_1$ for $\epsilon \in (0, 1/2)$. Then $W_2^2(\mu_{\epsilon}, \nu) = \epsilon$ (the optimal coupling moves mass ϵ from 0 to 1), while $\text{KL}(\mu_{\epsilon} \parallel \nu) = 2\epsilon^2 + O(\epsilon^4)$, so $W_2^2/\text{KL} = 1/(2\epsilon) + O(1) \rightarrow \infty$ as $\epsilon \rightarrow 0^+$ — no constant C uniformizes the ratio. The structural reason: Gozlan 2009^[23] established that T_2 is *equivalent* to dimension-free Gaussian concentration of $\mu^{\otimes n}$, strictly stronger than diameter (Talagrand-type product concentration requires LSI / curvature input, not just bounded support).

What does follow from diameter is the *weaker* T_1 inequality: by $W_1 \leq D \cdot \text{TV}$ and Pinsker $\text{TV} \leq \sqrt{\text{KL}/2}$,

$$W_1^2(P, P_{M_{\tau+|e}, M_{\tau-}}) \leq C_{T_1} \text{KL}(P \parallel P_{M_{\tau+|e}, M_{\tau-}}), \quad C_{T_1} = D^2/2$$

(squared convention matching (H2'); equivalently $W_1 \leq (D/\sqrt{2})\sqrt{\text{KL}}$, sharp on the two-point Bernoulli at $p = 1/2$ — Bobkov and Götze²⁸ with Hoeffding sub-Gaussian variance proxy $\sigma^2 = D^2/4$ for distributions on a diameter- D set, applied to the T_1 -equivalent Lipschitz-sub-Gaussian characterization). Substituting W_1 for W_2 in Step 2 of the Section 5.1 cascade preserves the *distance-form* bound $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq (D/\sqrt{2})\sqrt{I(G; M_{\tau+|e}, M_{\tau-})}$ at the T_1 level — at the cost of giving up Theorem 4.1's tighter squared-distance / linear-in- I refinement (5). Token-distribution belief states for LLMs on the simplex are more naturally treated under Track 2 (Fisher-Rao, Section 5.2) than via Track 1's diameter-only T_1 weakening; in finite-dimensional bounded-away-from-boundary simplex regimes, the universal $\sqrt{2}\sqrt{I}$ form holds.

Route (c): dimension-free sub-Gaussian Lipschitz concentration Per Gozlan 2009^[23], μ satisfies $T_2(C)$ iff $\mu^{\otimes n}$ enjoys dimension-free Gaussian concentration of Lipschitz functionals: there exist constants $C', \lambda > 0$ such that $\mathbb{E}_{\mu^{\otimes n}}[\exp(\lambda(f - \mathbb{E}f)^2)] \leq 2$ uniformly in n for every 1-Lipschitz $f : \mathbb{R}^n \rightarrow \mathbb{R}$. Bobkov and Götze 1999^[28] characterizes the strictly weaker *single-Lipschitz-function* version (giving T_1); the Gozlan strengthening to T_2 is what makes single-distribution sub-Gaussian moments insufficient on their own. When dimension-free product concentration holds, C_{T_2} admits an explicit form in the variance proxy.

When both routes fail Heavy-tailed post-updates without LSI and without dimension-free sub-Gaussian concentration are outside Track 1's T_2 scope. The T_1 fallback under bounded support (Route (b) as corrected above) recovers the distance form at weaker constants — adequate for $\mathbb{E}\|\Delta M_{\text{bias}}\| \leq C\sqrt{I}$. Track 2 under (PI) + (R) + (K) does not require (H2') and remains available; Theorem D.2's Hellinger backstop is a global universal-constant fallback under (PI) only.

672 B.2 KL-to-Fisher second-order expansion

673 For nearby distributions $P_\theta, P_{\theta+\delta}$ on a parametric family with Fisher information $\mathbf{I}(\theta)$:

$$\text{KL}(P_{\theta+\delta} \parallel P_\theta) = \frac{1}{2} \delta^\top \mathbf{I}(\theta) \delta + O(\|\delta\|^3) = \frac{1}{2} d_{FR}^2(P_\theta, P_{\theta+\delta}) + O(d_{FR}^3).$$

674 Standard derivation: Cover and Thomas²⁹ §12.5; Amari and Nagaoka¹⁹ §3.7 Theorem 3.1. The
675 $O(d_{FR}^3)$ remainder is sharp in the sense that the second-order term is exact and the third-order term
676 is generically non-zero (off-exponential-family).

677 B.3 $(\mathbf{H}_\kappa)$ sufficient conditions

678 The architectural-corollary commitment $(\mathbf{H}_\kappa)$ is that $\kappa_{\text{processing}} =$
679 $I(G; M_{\tau+} | e_\tau, M_{\tau-}) / I(G; \Omega_\tau | e_\tau, M_{\tau-})$ is bounded by some named architectural constant $\kappa^* < \infty$
680 (typically 1). For Class 1 (Separated) architectures, the data-processing inequality applied to the
681 Markov chain $G \rightarrow \Omega_\tau \rightarrow M_{\tau+}$ (conditional on $(e_\tau, M_{\tau-})$ by architectural separation) gives
682 $\kappa_{\text{processing}} \leq 1$ automatically. For Class 2 (Partial) and Class 3 (Coupled), two structural conditions
683 each give $\kappa_{\text{processing}} \leq 1$ (or $\leq 1 + \alpha$ in the second condition):

684 1. *Goal-blind effective kernel.* Suppose the post-update map factors as $M_{\tau+} =$
685 $K(M_{\tau-}, e_\tau, \tilde{\Omega}_\tau)$ with K deterministic, where $\tilde{\Omega}_\tau$ is a goal-conditional reweighting of
686 Ω_τ that is *non-amplifying*: $I(G; \tilde{\Omega}_\tau | e_\tau, M_{\tau-}) \leq I(G; \Omega_\tau | e_\tau, M_{\tau-})$. Then conditional
687 on $(e_\tau, M_{\tau-})$, $G \rightarrow \tilde{\Omega}_\tau \rightarrow M_{\tau+}$ is a Markov chain (since K depends on G only through
688 $\tilde{\Omega}_\tau$), and DPI applied along this chain gives

$$I(G; M_{\tau+} | e_\tau, M_{\tau-}) \leq I(G; \tilde{\Omega}_\tau | e_\tau, M_{\tau-}) \leq I(G; \Omega_\tau | e_\tau, M_{\tau-}),$$

689 i.e., $\kappa_{\text{processing}} \leq 1$.

690 1. *Bounded direct architectural channel.* Suppose the architecture’s *direct* goal-coupling
691 pathway (independent of Ω_τ) has bounded conditional MI relative to the resolvable chan-
692 nel: $I(G; M_{\tau+} | e_\tau, M_{\tau-}, \Omega_\tau) \leq \alpha I(G; \Omega_\tau | e_\tau, M_{\tau-})$ for some $\alpha \geq 0$. The chain
693 rule of mutual information conditioned on $(e_\tau, M_{\tau-})$ gives $I(G; M_{\tau+}, \Omega_\tau | e_\tau, M_{\tau-}) =$
694 $I(G; \Omega_\tau | e_\tau, M_{\tau-}) + I(G; M_{\tau+} | e_\tau, M_{\tau-}, \Omega_\tau)$, and using $I(G; M_{\tau+} | e_\tau, M_{\tau-}) \leq$
695 $I(G; M_{\tau+}, \Omega_\tau | e_\tau, M_{\tau-})$ (additional side information cannot decrease MI), we get
696 $\kappa_{\text{processing}} \leq 1 + \alpha$. The case $\alpha = 0$ — architectural channel adds nothing beyond what Ω_τ
697 provides given $(e_\tau, M_{\tau-})$ — gives $\kappa_{\text{processing}} \leq 1$.

698 **Worst-case witness — parrot architecture** Consider a degenerate Coupled-class architecture
699 whose update rule is $M_{\tau+} := G$ (the post-update model state is identically the goal). Two computa-
700 tions:

701 *Numerator.* Since $M_{\tau+} = G$ deterministically conditional on $(e_\tau, M_{\tau-})$, $I(G; M_{\tau+} | e_\tau, M_{\tau-}) =$
702 $H(G | e_\tau, M_{\tau-})$ exactly.

703 *Denominator.* Suppose $G \perp\!\!\!\perp \Omega_\tau$ given $(e_\tau, M_{\tau-})$ — the goal carries no information about the latent
704 world-state beyond what the evidence and prior have already pinned down. (Operationally this is the
705 “make me feel good” goal of Section 3.2: the user’s affective goal is independent of the empirical
706 question the evidence resolves.) Then $I(G; \Omega_\tau | e_\tau, M_{\tau-}) = 0$.

707 Hence $\kappa_{\text{processing}} = H(G | e_\tau, M_{\tau-}) / 0 = +\infty$. $(\mathbf{H}_\kappa)$ explicitly fails. This shows the architectural
708 commitment is *real* for non-Separated architectures, not a free move. The unconditional theorems
709 (Theorem 4.1, Theorem D.1, Theorem D.2) still bound the parrot architecture’s displacement by
710 transferred information directly — the prefactor is $\sqrt{H(G | e_\tau, M_{\tau-})}$ rather than $\sqrt{\kappa I(G; \Omega)}$ — but
711 the $\kappa \times \mathcal{A}$ factorization (and hence the architectural-corollary form) does not apply.

C Conjugate-Gaussian numerical comparison

We compute the Track-family constants explicitly for the conjugate-Gaussian observation-prior pair and exhibit the regimes in which they are tight. *The setup separates the agent's internal model from the operating distribution that probes goal-coupling.* The agent uses the conjugate Gaussian prior/likelihood update rule with prior $\theta \sim \mathcal{N}(0, \tau^2)$ and observation channel $\Omega|\theta \sim \mathcal{N}(\theta, \sigma^2)$. The external operating distribution used to probe goal-coupling has $\Omega|G = g \sim \mathcal{N}(\beta(g), \sigma^2)$ — the goal directly shifts the operating-data mean. The architecture's update is goal-blind (Class 1 Separated, conjugate Bayesian update). Identifying $M_{\tau+}$ with the posterior-mean parameter $\mu_+ := L_{\text{post}}\Omega$ (variance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ fixed), the post-update parameter law under operating G is $\mu_+|G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$.

Two distinct variances. The *inside-posterior covariance* $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ is the spread of the posterior measure on θ at a fixed observation. The *parameter-law variance* $L_{\text{post}}^2\sigma^2 = L_{\text{post}}\sigma_{\text{post}}^2$ is the spread of the random model state μ_+ under data noise. They differ by exactly L_{post} . Track 1's C_{T_2} is determined by the parameter-law variance, not the inside-posterior covariance; conflating them is the algebraic error this appendix corrects.

Transferred information and $\kappa_{\text{processing}}$. The goal-conditional law of Ω has small- $\text{Var}_G(\beta)$ MI $I(G; \Omega|e, M_{\tau-}) \approx \text{Var}_G(\beta)/(2\sigma^2)$. The post-update parameter law inherits the same MI: the deterministic linear map $\Omega \mapsto L_{\text{post}}\Omega$ is invertible, so the data-processing inequality on the chain $G \rightarrow \Omega \rightarrow M_{\tau+}$ saturates and $I(G; M_{\tau+}|e, M_{\tau-}) = I(G; \Omega|e, M_{\tau-})$ exactly. The realized attenuation ratio is $\kappa_{\text{processing}}^{\text{realized}} = 1$. *This Class 1 conjugate-Gaussian setup does not exhibit architectural attenuation* — it sits at the worst-case Class 1 corner where DPI is tight; non-trivial $\kappa < 1$ requires a non-invertible data channel (e.g., quantization, dimension reduction) or a Class 2/3 architecture.

Track 1 (Theorem 4.1). Route (a) — LSI on the post-update parameter law $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$ at small $\text{Var}_G(\beta)$ — gives $\rho_{\text{post}} = 1/(L_{\text{post}}^2\sigma^2)$ and

$$C_{T_2} = 2/\rho_{\text{post}} = 2L_{\text{post}}^2\sigma^2 = \frac{2\tau^4\sigma^2}{(\sigma^2 + \tau^2)^2}.$$

Theorem 4.1: $\mathbb{E}[W_2^2] \leq C_{T_2} \cdot I(G; M_{\tau+}|e, M_{\tau-}) = 2L_{\text{post}}^2\sigma^2 \cdot I(G; \Omega|e, M_{\tau-})$ (using $\kappa = 1$). Equivalently, identifying $\rho_{\text{noise}} := 1/\sigma^2$ as the *noise concentration* of the conditional data law $\Omega | \theta$ (the quantity that enters the Stuart-school cascade's denominator — distinct from the LSI constant of the marginal observation law $P_\Omega \sim \mathcal{N}(0, \sigma^2 + \tau^2)$, which would be $1/(\sigma^2 + \tau^2)$),

$$\mathbb{E}[W_2^2] \leq \frac{2L_{\text{post}}^2}{\rho_{\text{noise}}} \cdot I(G; \Omega|e, M_{\tau-})$$

— the canonical Stuart-school cascade constant, recovered as C_{T_2} alone (DPI saturation $\kappa = 1$ is the witness that this Class 1 example sits at the Stuart corner). The constant $2L_{\text{post}}^2\sigma^2$ here is read off *directly* from the post-update parameter law's variance $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$ — sub-Gaussian by inspection. The general structural reduction Theorem E.3 across the broader Stuart-school hypothesis space recovers a constant of the same canonical form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$, with an absolute multiplicative overhead from the kernel-tensorization step; the conjugate-Gaussian instance lands at the *tight* constant via the direct read-off rather than via the abstract reduction. The cascade is sharp here: the slice-wise W_2^2/KL ratio equals C_{T_2} exactly for these Gaussian parameter laws. Limit checks: prior-dominant ($\tau \ll \sigma$): $C_{T_2} \approx 2\tau^4/\sigma^2 \rightarrow 0$; likelihood-dominant ($\sigma \ll \tau$): $C_{T_2} \approx 2\sigma^2 \rightarrow 0$; balanced ($\tau = \sigma = 1$): $C_{T_2} = 1/2$ and $\mathbb{E}[W_2^2] \leq I_M/2$.

Track 2 (Theorem 4.1). Fisher metric on the parameter-law manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ (laws over the random parameter μ_+ , not posterior measures on θ): $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$. Fisher-Rao distance $d_{FR}(\mathcal{N}(\mu_1, v), \mathcal{N}(\mu_2, v)) = |\mu_1 - \mu_2|/\sqrt{v} = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$. Slice-wise to the marginal $\mathcal{N}(0, L_{\text{post}}^2\sigma^2)$: $d_{FR}(P_{M_{\tau+}|G=g}, P_{M_{\tau+}}) \approx |\beta(g)|/\sigma$ at small $\text{Var}_G(\beta)$. Theorem 4.1 (unconditional): $\mathbb{E}[d_{FR}^2] \leq 2 \cdot I(G; M_{\tau+}|e, M_{\tau-}) = \text{Var}_G(\beta)/\sigma^2$ — sharp on this family.

Track 2 Euclidean translation (Theorem D.4). $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$, so $\mathbb{E}|\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}}\sigma\sqrt{2I_M}$. The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM

at $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the architecture learns essentially nothing from Ω , posterior-mean displacement vanishes); $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow 0$ (sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ” reading is wrong (correct prefactor is $L_{\text{post}}\sigma$, not bare σ); equally importantly, the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading is *also* wrong (it conflates inside-posterior covariance $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$).

Hellinger backstop (Theorem D.2). Theorem D.2: $\mathbb{E} \text{Hel}(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq (1/\sqrt{2})\sqrt{I_M}$ globally, where Hel is between two parameter laws $\mathcal{N}(\mu_1, L_{\text{post}}^2\sigma^2)$ and $\mathcal{N}(\mu_2, L_{\text{post}}^2\sigma^2)$. Hellinger is globally bounded by 1, so the bound is non-trivial when $\sqrt{I_M}/2 < 1$ — i.e., $I_M < 2$ nat.

Numerical comparison at balanced scale ($\tau = \sigma = 1$, $L_{\text{post}} = 1/2$, $L_{\text{post}}\sigma = 1/2$, $L_{\text{post}}^2\sigma^2 = 1/4$, transferred info $I_M = 1$ nat for normalization):

Table 2: Conjugate-Gaussian numerical comparison at balanced scale ($\tau = \sigma = 1$, $I_M = 1$ nat).

Track	Theorem	Bound	Numerical at $I_M = 1$
Track 1	Theorem 4.1	$\mathbb{E}[W_2^2] \leq C_{T_2} I_M = (1/2) I_M$	$\mathbb{E} W_2 \leq 1/\sqrt{2}$
Track 2	Theorem 4.1	$\mathbb{E} d_{FR} \leq \sqrt{2I_M}$	$\mathbb{E} d_{FR} \leq \sqrt{2}$
Track 2 Euclidean translation	Theorem D.4	$\mathbb{E} \ \Delta\mu_+\ \leq L_{\text{post}}\sigma\sqrt{2I_M} = (1/2)\sqrt{2I_M}$	$\mathbb{E} \ \Delta\mu_+\ \leq 1/\sqrt{2}$
Hellinger	Theorem D.2	$\mathbb{E} \text{Hel} \leq \sqrt{I_M}/2$	$\mathbb{E} \text{Hel} \leq 1/\sqrt{2}$

Track 1 and the Track 2 Euclidean translation agree exactly at balanced scale ($\mathbb{E} W_2 = \mathbb{E} \|\Delta\mu_+\|_{\text{Eucl}} = 1/\sqrt{2}$ at $I_M = 1$) — they bound the same Euclidean displacement; the Track 2 Euclidean translation is Track 2 with the Fisher-Rao-to-Euclidean conversion factor $L_{\text{post}}\sigma$ on the parameter chart. The Euclidean translation goes to zero in both extreme limits (vanishing prior-dominant and vanishing likelihood-dominant). At $I_M = 1$ nat the Hellinger bound saturates the global $\sqrt{1/2} \approx 0.71$; for $I_M \ll 1$ Track 1 / Track 2 Euclidean are tighter; for $I_M \gtrsim 2$ the Hellinger backstop is the only non-vacuous bound.

D Track 2 companions and parametric Euclidean translations

The locally-tight Fisher-Rao geodesic bound (Theorem 4.1’s Track 2 instantiation under (PI)+(R)+(K)+(H4’)) is one of three Fisher-Rao-spine bounds; the other two — a globally-valid *ambient Fisher-Rao spherical-arc* bound and a Hellinger chord backstop — recover under strictly weaker hypotheses, with universal constants of their own. Together with the conjugate-Gaussian Euclidean translation (clarifying the no-go’s apparent scale-family pathology) and the exponential-family Euclidean generalization, they form the Track 2 family.

D.1 Track 2 global Fisher-Rao spherical-arc backstop

Outside the uniform-locality regime — when a few rare goals carry disproportionately large transferred information, or when the operating distribution sits at moderate-to-large I_M — the slice-wise expansion’s $(1 + o(1))$ remainder is no longer uniformly sharp. A global *ambient spherical-arc* bound is available under strictly weaker hypotheses, at the cost of a slightly larger constant. (The bound is on the Bhattacharyya / Fisher spherical-arc pseudometric $d_{FR} = 2 \arccos(\text{BC})$ on the unit L^2 -sphere of $\sqrt{p}$, which equals intrinsic Fisher-Rao only on the ambient simplex / full nonparametric distribution space; on parametric submanifolds it is the chart-invariant pseudometric, $\leq$ submanifold-intrinsic geodesics.)

Theorem D.1 (Track 2 global Fisher-Rao spherical-arc backstop). *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E} d_{FR}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq 2 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

The constant 2 is universal, dimension-free, global — no small-information condition required, no (R), no (K). Tightest form: $\mathbb{E} d_{FR} \leq 2 \arccos(\exp(-I_M/2)) \leq \min(2\sqrt{I_M}, \pi)$, by Jensen on the concave map $\psi(K) := 2 \arccos(\exp(-K/2))$ — ψ is concave on $[0, \infty)$ (verifiable by $\psi'(K) =$

exp($-K/2$)/ $\sqrt{1 - \exp(-K)}$ decreasing). The constant 2 is sharp — approached arbitrarily closely as $N \rightarrow \infty$ on a symmetric N -point family (Section D.1.1 below).

Proof. By Rényi-divergence monotonicity^[30], $D_{1/2}(P\|Q) \leq D_1(P\|Q) = \text{KL}(P\|Q)$, equivalently $\text{BC}(P, Q) := \int \sqrt{pq} \geq \exp(-\text{KL}/2)$. Combined with the chord-arc identity (not inequality) $d_{FR} = 2 \arccos(\text{BC})$ from the unit- L^2 -sphere geometry: $d_{FR} \leq 2 \arccos(\exp(-\text{KL}/2))$. Define $\phi(K) := 4 \arccos^2(\exp(-K/2))$; series expansion at $K = 0$ gives $\phi(K)/K \rightarrow 4$, decreasing thereafter, so $\phi(K) \leq 4K$ globally. Hence $d_{FR}^2 \leq 4 \text{KL}$ slice-wise. Take $\mathbb{E}_G$, substitute the chain rule Lemma 5.1: $\mathbb{E} d_{FR}^2 \leq 4I_M$. Jensen on $\sqrt{x}$: $\mathbb{E} d_{FR} \leq 2\sqrt{I_M}$. The tightest form uses concavity of ϕ before Jensen on $\sqrt{x}$. $\square$

The $\sqrt{2}$ overhead vs. the locally-tight $\sqrt{2}$ — i.e., $2/\sqrt{2} = \sqrt{2}$ in raw constant ratio — is *exactly* the chord-arc factor at the unit L^2 -sphere’s antipode in squared-distance form. Under (PI) alone, the Bhattacharyya-coefficient pseudometric $d_{FR} = 2 \arccos(\text{BC})$ is automatically (PI)-invariant (BC is an f -divergence), and the bound $d_{FR} \leq 2\sqrt{\text{KL}}$ is the cleanest possible (PI)-invariant Fisher-Rao-to-KL bound; (R) + (K) add the local refinement to $\sqrt{2}$. This composition (exact chord-arc identity + Rényi-1/2 monotonicity) is strictly tighter than the (chord-arc inequality + Tsybakov 2.4) composition by factor $\pi/(2\sqrt{2}) \approx 1.11$ — Tsybakov 2.4 is loose by factor 2 at small KL ($\text{Hel}^2 \approx \text{KL}/4$ locally vs Tsybakov’s $\text{Hel}^2 \leq \text{KL}/2$), and Rényi-1/2 monotonicity recovers the lost factor.

D.1.1 Sharpness witness

The constant 2 is sharp via a symmetric N -point construction. Take G uniform on $\{1, \dots, N\}$ with $P_{M_{\tau+}|G=g}$ uniform on the $N-1$ atoms other than g . By symmetry $P_{M_{\tau+}}$ is uniform on all N atoms, so $\text{KL}(P_{M_{\tau+}|G=g}\|P_{M_{\tau+}}) = \log \frac{N}{N-1}$ (constant in g , so $I_M = \log \frac{N}{N-1}$) and $\text{BC} = \sqrt{(N-1)/N}$, giving $d_{FR} = 2 \arcsin(1/\sqrt{N})$ (also constant in g , so $\text{Var}_G(d_{FR}) = 0$ and Jensen is tight). The ratio $\mathbb{E} d_{FR}/\sqrt{I_M} = 2 \arcsin(1/\sqrt{N})/\sqrt{\log \frac{N}{N-1}} \rightarrow 2$ at rate $1 - 1/(12N) + O(N^{-2})$ as $N \rightarrow \infty$. Numerical confirmation: $N = 100$ achieves 99.92% saturation, $N = 1000$ achieves 99.992%. The discrete-alphabet structure — large vocabulary $|V|$ admitting near-disjoint goal-conditional supports — is exactly the structural shape adversarial / jailbreak prompts exploit; the witness is the natural worst-case for goal-coupled architectures.

D.2 Hellinger backstop

The chord companion to Theorem D.1’s arc bound, with the cleanest universal constant.

Theorem D.2 (Hellinger backstop). *Under (H1) and (PI) — for all values of transferred information,*

$$\mathbb{E} \text{Hel}(P_{M_{\tau+}|e, M_{\tau-}, G}, P_{M_{\tau+}|e, M_{\tau-}}) \leq (1/\sqrt{2}) \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

Under (H_{κ}) , the architectural factorization recovers: $\mathbb{E} \text{Hel} \leq (1/\sqrt{2})\sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega | e, M)}$.

Proof. By Tsybakov³¹, Lemma 2.4, $2\text{Hel}^2(P, Q) \leq \text{KL}(P\|Q)$ globally for all probability measures. Apply slice-wise at each $G = g$, take expectation over G , and substitute the chain rule Lemma 5.1. $\square$

Hellinger satisfies (PI) — it is an f -divergence with $f(t) = (\sqrt{t} - 1)^2/2$ preserved under Markov morphisms^[32,33] — and is locally proportional to Fisher-Rao on the unit-sphere representation. Throughout we adopt the *standard statistician’s* Hellinger convention $\text{Hel}^2(P, Q) := \frac{1}{2}\|\sqrt{p} - \sqrt{q}\|_{L^2}^2 = 1 - \int \sqrt{pq}$, so $\text{Hel} \in [0, 1]$, and the *Amari-Nagaoka* Fisher-Rao convention $d_{FR}(P, Q) = 2 \arccos \int \sqrt{pq}$ (twice the great-circle arc on the unit sphere of $\sqrt{p}$ in L^2 , $d_{FR} \in [0, \pi]$). Different conventions in the literature differ by factors of 2; the forms above are the ones consistent with $\text{Hel} \in [0, 1]$, $d_{FR} \in [0, \pi]$, and the $2\text{Hel}^2 \leq \text{KL}$ inequality used in the proof.

The three Fisher-Rao-spine bounds — locally-tight $\sqrt{2}$ (under (H1)+(H4’)+(PI)+(R)+(K)), globally-valid 2 (Theorem D.1, under (H1)+(PI) only), and Hellinger chord $1/\sqrt{2}$ (Theorem D.2, under (H1)+(PI) only) — all operate on the unit L^2 -sphere of $\sqrt{p}$ via the Amari-Nagaoka spherical-arc

convention adopted in Section 3.1. The local theorem under (PI)+(R)+(K)+(H4') coincides with intrinsic Fisher-Rao on parametric submanifolds at second order; the global theorems under (PI) alone bound the *ambient* spherical-arc pseudometric and apply to any pair of probability distributions regardless of submanifold structure. The hypothesis sets are strictly nested and the bounds use the same chord-arc geometry of the L^2 -sphere; the global constants (2 for the arc, $1/\sqrt{2}$ for the chord) are companion bounds at the antipode.

D.3 Hellinger-Fisher-Rao chord-arc geometry

Lemma D.3 (Global chord-arc bound on the unit L^2 -sphere). *For any probability measures P, Q , $d_{FR}(P, Q) \leq \pi \cdot \text{Hel}(P, Q)$, with equality at the antipode $\text{Hel}(P, Q) = 1$ (correspondingly $d_{FR}(P, Q) = \pi$).*

Proof. From $d_{FR} = 4 \arcsin(\text{Hel}/\sqrt{2})$, define $\phi : (0, 1] \rightarrow (0, \pi]$ by $\phi(h) := 4 \arcsin(h/\sqrt{2})/h$, so $d_{FR}/\text{Hel} = \phi(\text{Hel})$. Set $g(h) := 4 \arcsin(h/\sqrt{2})$. Differentiating: $g'(h) = 4/\sqrt{2-h^2}$ for $h \in [0, \sqrt{2})$, $g''(h) = 4h/(2-h^2)^{3/2} > 0$ on $(0, \sqrt{2})$ — so g is *strictly convex* on this interval, with $g(0) = 0$. By the secant-slope monotonicity for convex functions vanishing at zero, $\phi(h) = g(h)/h$ is monotonically increasing on $(0, \sqrt{2})$, with limits $\lim_{h \rightarrow 0^+} \phi(h) = g'(0) = 4/\sqrt{2} = 2\sqrt{2}$ and $\phi(1) = g(1) = 4 \arcsin(1/\sqrt{2}) = \pi$. Hence $\phi(h) \leq \pi$ for all $h \in (0, 1]$, with equality at $h = 1$. $\square$

Local proportionality As $d_{FR} \rightarrow 0$, $\text{Hel} \approx d_{FR}/(2\sqrt{2})$, so Theorem 4.1's Track 2 locally-tight bound $\mathbb{E} d_{FR} \leq \sqrt{2I_M}$ jointly with Lemma D.3 gives $\mathbb{E} \text{Hel} \lesssim (1/2)\sqrt{I_M}$ in the small-information limit — *tighter* than Theorem D.2's global $(1/\sqrt{2})\sqrt{I_M}$ by a factor of $\sqrt{2}$. So in the (H4') regime, Theorem 4.1 + Lemma D.3 jointly give the tightest Hellinger bound; Theorem D.2 (under (PI) only, no (H4')) is the global statement.

D.4 Conjugate-Gaussian Euclidean translation

When (H1) is realized by a parametric family with known Fisher information, the locally-tight Fisher-Rao bound translates to an explicit Euclidean bound on parameter displacement. The conjugate-Gaussian case clarifies the no-go's scope: *Euclidean displacement does not blow up* in the prior-dominant limit, contrary to a naive reading of the chart-rescaling argument.

Theorem D.4 (Conjugate-Gaussian Euclidean bound). *As in Appendix C, the agent uses the conjugate Gaussian prior/likelihood update rule ($\theta \sim \mathcal{N}(0, \tau^2)$, $\Omega \mid \theta \sim \mathcal{N}(\theta, \sigma^2)$), while the operating distribution probing goal-coupling has $\Omega \mid G = g \sim \mathcal{N}(\beta(g), \sigma^2)$. The post-update parameter law under operating G is the Gaussian $\mu_+ \mid G \sim \mathcal{N}(L_{\text{post}}\beta(G), L_{\text{post}}^2\sigma^2)$ with $L_{\text{post}} = \tau^2/(\sigma^2 + \tau^2)$. The locally-tight Fisher-Rao bound translates to the Euclidean bound on random-parameter displacement*

$$\mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}} \sigma \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

with prefactor $L_{\text{post}} \sigma = \tau^2 \sigma / (\sigma^2 + \tau^2) \leq \tau/2$ uniformly. Under (H_κ) , $\mathbb{E} |\Delta\mu_+|_{\text{Eucl}} \leq L_{\text{post}} \sigma \sqrt{2\kappa_{\text{processing}} I(G; \Omega \mid e, M)} (1 + o(1))$.

Proof. The post-update parameter law $\mu_+ \mid G$ is Gaussian with mean $L_{\text{post}}\beta(G)$ and variance $L_{\text{post}}^2\sigma^2$ (variance preserved across goals because σ_{post}^2 is fixed by the conjugate-update structure). The Fisher metric on the manifold $\{\mathcal{N}(\mu, L_{\text{post}}^2\sigma^2) : \mu \in \mathbb{R}\}$ in the μ -coordinate is $\mathbf{I}(\mu) = 1/(L_{\text{post}}^2\sigma^2)$, so the Fisher-Rao geodesic distance between two such laws is $d_{FR}(\mu_1, \mu_2) = |\mu_1 - \mu_2|/(L_{\text{post}}\sigma)$ — a flat metric pulled back from $\mathbb{R}$. Hence $|\Delta\mu_+|_{\text{Eucl}} = L_{\text{post}}\sigma \cdot d_{FR}$. Take expectations and substitute the locally-tight Fisher-Rao bound. $\square$

The prefactor $L_{\text{post}}\sigma = \tau^2\sigma/(\sigma^2 + \tau^2)$ is bounded above by $\tau/2$ uniformly (AM-GM at the maximizer $\sigma = \tau$) and *vanishes in both extreme limits*: $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow \infty$ (very noisy observations: the architecture learns essentially nothing from Ω regardless of goal) and $L_{\text{post}}\sigma \rightarrow 0$ as $\sigma \rightarrow 0$ (very sharp observations: $L_{\text{post}} \rightarrow 1$ but $\sigma \rightarrow 0$, so even though the posterior tracks data tightly the data variance vanishes). Maximum at balanced $\sigma = \tau$. The naive “ $\sigma\sqrt{2I}$ ” reading is wrong (correct prefactor is $L_{\text{post}}\sigma$); the “ $\sigma_{\text{post}}\sqrt{2I}$ ” reading is *also* wrong (it conflates inside-posterior covariance

889 $\sigma_{\text{post}}^2 = L_{\text{post}}\sigma^2$ with parameter-law variance $L_{\text{post}}^2\sigma^2$). The Section 4.1 no-go still applies as stated to
 890 *charts*: $L_{\text{post}}\sigma$ is family-specific.

891 D.5 Exponential-family Euclidean generalization

892 For an exponential family $P_\theta(x) = h(x) \exp(\theta^\top T(x) - A(\theta))$ with natural parameter $\theta \in \Theta \subseteq \mathbb{R}^d$
 893 and log-partition A , the Fisher information metric is $\mathbf{I}(\theta) = \nabla^2 A(\theta)$. The infinitesimal Fisher-Rao
 894 expansion gives $d_{FR}^2(P_{\theta_1}, P_{\theta_2}) = (\theta_1 - \theta_2)^\top \mathbf{I}(\theta^*)(\theta_1 - \theta_2) + O(\|\theta_1 - \theta_2\|^3)$ for θ^* on the geodesic
 895 between θ_1, θ_2 . So Euclidean displacement on the natural parameter satisfies $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} \cdot d_{FR}^2$
 896 where $\mathbf{I}_{\min}$ is the minimum eigenvalue of $\mathbf{I}$ along the geodesic.

897 **Theorem D.5** (Exponential-family Euclidean bound). *For an exponential family in natural-parameter*
 898 *coordinates with Fisher-information minimum-eigenvalue $\mathbf{I}_{\min}$ along the goal-induced geodesic, the*
 899 *locally-tight Fisher-Rao bound translates to*

$$900 \mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \cdot \sqrt{2I_M} \cdot (1 + o(1)),$$

$$901 \text{ and under } (H_\kappa): \mathbb{E} \|\Delta\theta_{\text{bias}}\|_{\text{Eucl}} \leq \mathbf{I}_{\min}^{-1/2} \sqrt{2\kappa_{\text{processing}} I(G; \Omega \mid e, M)} (1 + o(1)).$$

902 *Proof.* By Theorem 4.1's Track 2 instantiation, $\mathbb{E} d_{FR}^2 \leq 2I_M(1 + o(1))$. By the infinitesimal
 903 Fisher-Rao expansion, $\|\theta_1 - \theta_2\|^2 \leq \mathbf{I}_{\min}^{-1} d_{FR}^2$. Take expectations and Jensen. $\square$

904 **Worked examples** *Bernoulli on logit chart, restricted to $p \in [\delta, 1 - \delta]$: $\mathbf{I}(\theta) = p(1 - p)$ has no*
 905 *positive lower bound on the full chart $\theta \in \mathbb{R}$; restricting to $p \in [\delta, 1 - \delta]$ gives $\mathbf{I}_{\min} \geq \delta(1 - \delta)$*
 906 *and finite, dimension-free Euclidean bound $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq [\delta(1 - \delta)]^{-1/2} \sqrt{2I_M}$ — contingent on*
 907 *boundary avoidance. Multinomial on log-ratio chart: Fisher minimum eigenvalue bounded below*
 908 *by $\min_i p_i$ on the simplex interior; for δ -bounded models, $\mathbb{E} \|\Delta\theta\|_{\text{Eucl}} \leq \delta^{-1/2} \sqrt{2I_M}$. Gaussian-*
 909 *mean chart with fixed variance v : $\mathbf{I}(\mu) = 1/v$ constant on the manifold; $\mathbb{E} \|\Delta\mu\|_{\text{Eucl}} \leq \sqrt{v} \sqrt{2I_M}$.*
 910 *Gaussian-scale chart $\theta = \sigma$: $\mathbf{I}(\sigma) = 2/\sigma^2$ — $\mathbf{I}_{\min}^{-1/2} = \sigma/\sqrt{2}$ grows with σ , recovering the no-go's*
 911 *pathology consistent with Section 4.1; reparameterizing to $\theta = \log \sigma$ gives $\mathbf{I}(\log \sigma) = 2$ constant,*
 912 *restoring a finite Euclidean bound.*

913 The pattern: for natural-parameter charts on simplex / discrete-outcome models with the goal-induced
 914 geodesic restricted away from the boundary, the Euclidean Track 2 translation is dimension-free and
 915 finite; for scale-family parametrizations off the natural-parameter chart, the prefactor grows in σ or
 916 λ^{-1} — recovering Section 4.1's pathology consistent with the no-go.

917 E Proofs of main results

918 This appendix collects full proofs deferred from §4–§5 of the main text. Proof structure: each section
 919 restates the formal claim from main text, then provides the full derivation. Auxiliary lemmas used in
 920 proofs are stated where they first arise.

921 E.1 Track 1 cascade — full derivation

922 The umbrella theorem's Track 1 instantiation (Theorem 4.1 under (H1) + (H2')) follows from a
 923 two-step composition: chain rule on the post-update law, then slice-wise Talagrand T_2 .

924 *Proof. Step 1 — Chain rule of relative entropy on the post-update law.* By
 925 Cover and Thomas 2006^[29] Theorem 2.5.3 applied to the conditional law of $M_{\tau+}$ given $(e_\tau, M_{\tau-})$
 926 marginalized over G :

$$927 \mathbb{E}_G [\text{KL}(P_{M_{\tau+} | e_\tau, M_{\tau-}, G} \| P_{M_{\tau+} | e_\tau, M_{\tau-}})] = I(G; M_{\tau+} \mid e_\tau, M_{\tau-}).$$

928 The (H1) standard-Borel regularity makes this hold in the abstract-spaces form (Polyanskiy and Wu²¹,
 929 Theorem 3.4; Gray²², Theorem 5.4). The right-hand side is *transferred* goal-information into the
 930 post-update model state.

931 *Step 2 — Slice-wise Talagrand T_2 .* Apply Hypothesis 4.2 slice-wise at each $G = g$:

$$W_2^2(P_{M_{\tau+}|e,M,G=g}, P_{M_{\tau+}|e,M}) \leq C_{T_2} \cdot \text{KL}(P_{M_{\tau+}|e,M,G=g} \parallel P_{M_{\tau+}|e,M}).$$

Take expectation over G and substitute Step 1:

$$\mathbb{E}[W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq C_{T_2} \cdot I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-}).$$

Jensen's inequality on $\sqrt{x}$ delivers the distance form

$$\mathbb{E} W_2(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq \sqrt{C_{T_2}} \cdot \sqrt{I(G; M_{\tau+} \mid e_{\tau}, M_{\tau-})}. \quad \square$$

The architectural-corollary form (Corollary 4.7 under (H_{κ})) follows by substituting $I(G; M_{\tau+} \mid e, M_{\tau-}) \leq \kappa_{\text{processing}} \cdot I(G; \Omega_{\tau} \mid e, M_{\tau-})$ on the right-hand side; the substitution is *definitional* per (4), with the boundedness of $\kappa_{\text{processing}}$ being the substantive content of (H_{κ}) .

E.2 Lipschitz pushforward of T_2 — Stuart-school reduction

Track 1's $(H2')$ is most naturally verified through a two-step Lipschitz-pushforward of T_2 that matches the canonical Stuart-school cascade across the full Strand 2 hypothesis space — not only conjugate-Gaussian. The reduction is one textbook lemma + one elementary composition.

Lemma E.1 (Lipschitz pushforward of T_2). *Let (X, d_X) and (Y, d_Y) be Polish metric spaces, μ on X satisfying Talagrand $T_2(C)$ — that is, $W_2^{d_X}(\nu, \mu)^2 \leq C \text{KL}(\nu \parallel \mu)$ for all $\nu \ll \mu$ — and let $T : X \rightarrow Y$ be L -Lipschitz. Then $T_*\mu$ on Y satisfies $T_2(L^2C)$.*

Proof. Let $\rho \ll T_*\mu$. Disintegrate $\mu(dx) = \mu_y(dx) T_*\mu(dy)$ along T . The lift $\tilde{\rho}(dx) := \mu_y(dx) \rho(dy)$ satisfies $T_*\tilde{\rho} = \rho$ and $\text{KL}(\tilde{\rho} \parallel \mu) = \text{KL}(\rho \parallel T_*\mu)$ (chain-rule decomposition with shared conditionals). For any coupling π of $(\tilde{\rho}, \mu)$, the pushforward $(T \otimes T)_*\pi$ couples $(\rho, T_*\mu)$ with cost $\leq L^2$ times the input cost. Hence $W_2^{d_Y}(\rho, T_*\mu)^2 \leq L^2 W_2^{d_X}(\tilde{\rho}, \mu)^2 \leq L^2 C \text{KL}(\tilde{\rho} \parallel \mu) = L^2 C \text{KL}(\rho \parallel T_*\mu)$. (Standard T_2 -tensorization-style argument; cf. Bobkov and Götze²⁸.) $\square$

The reduction needs an additional tensorization lemma — T_2 on the goal-induced parameter measure tensorizes through a Lipschitz-conditional kernel into T_2 on the joint, then projects to the marginal observation law, then pushes forward through the Bayesian-update map.

Lemma E.2 (Tensorization of T_2 with Lipschitz-conditional kernel). *Let P_{θ} on Θ satisfy Talagrand $T_2(C_{\theta})$, and let kernel $K(\theta, \cdot) := P_{\Omega|\theta}$ satisfy (i) $K(\theta, \cdot) \in T_2(C_{\text{slice}})$ uniformly in θ , and (ii) $W_2(K(\theta_1, \cdot), K(\theta_2, \cdot)) \leq L_{\text{lik}} \|\theta_1 - \theta_2\|$. Then the joint $P_{\theta} \otimes K$ on $\Theta \times \Omega$ with the L^2 -product metric satisfies $T_2(C_J)$ with $C_J \leq \max((1 + 2L_{\text{lik}}^2) C_{\theta}, 2C_{\text{slice}})$.*

Proof. Disintegrate $\rho \ll P_{\theta} \otimes K$ as $\rho(d\theta, d\Omega) = \rho_{\theta}(d\Omega) \rho_{\Theta}(d\theta)$. Chain rule on KL: $\text{KL}(\rho \parallel P_{\theta} \otimes K) = \text{KL}(\rho_{\Theta} \parallel P_{\theta}) + \int \text{KL}(\rho_{\Omega} \parallel K(\theta, \cdot)) d\rho_{\Theta}$. Construct a coupling: π^{Θ} optimal W_2 between $(\rho_{\Theta}, P_{\theta})$; for each (θ, θ') , $\pi^{\Omega}(\theta, \theta')$ optimal W_2 between $(\rho_{\theta}, K(\theta', \cdot))$. The product coupling has marginals ρ and $P_{\theta} \otimes K$, with L^2 -product cost $W_2^2(\rho_{\Theta}, P_{\theta}) + \iint W_2^2(\rho_{\theta}, K(\theta', \cdot)) d\pi^{\Theta}$. Triangle + Cauchy-Schwarz with (ii): $W_2^2(\rho_{\theta}, K(\theta', \cdot)) \leq 2W_2^2(\rho_{\theta}, K(\theta, \cdot)) + 2L_{\text{lik}}^2 \|\theta - \theta'\|^2$. Apply (i) to the first piece, $T_2(C_{\theta})$ to the marginal, combine: $W_2^2(\rho, P_{\theta} \otimes K) \leq (1 + 2L_{\text{lik}}^2) C_{\theta} \text{KL}(\rho_{\Theta} \parallel P_{\theta}) + 2C_{\text{slice}} \int \text{KL}(\rho_{\Omega} \parallel K(\theta, \cdot)) d\rho_{\Theta} \leq \max((1 + 2L_{\text{lik}}^2) C_{\theta}, 2C_{\text{slice}}) \text{KL}(\rho \parallel P_{\theta} \otimes K)$. $\square$

Theorem E.3 (Stuart-school reduction — generalized cascade-constant transfer). *Suppose:*

(C0) (Goal-induced parameter concentration.) $P_{\theta} := \theta_* P_G$ satisfies Talagrand $T_2(C_{\theta})$ on parameter space.

(C1) (Class 1 architecture.) $G \rightarrow \Omega_{\tau} \rightarrow M_{\tau+}$ is a Markov chain conditional on $(e_{\tau}, M_{\tau-})$.

(C2) (Deterministic Bayesian update.) $M_{\tau+} = \mu^{\Omega_{\tau}}$ for a measurable map $\Omega \mapsto \mu^{\Omega}$ on $\text{supp } P_{\Omega}$.

(C3) (Stuart-school Lipschitz posterior.) $W_2(\mu^{\Omega}, \mu^{\Omega'}) \leq L_{\text{post}} \|\Omega - \Omega'\|$ uniformly on $\text{supp } P_{\Omega}$.

(C4a) (Slice T_2 .) For P_{θ} -a.e. θ , $P_{\Omega|\theta} \in T_2(2/\rho_{\text{noise}})$.

(C4b) (Lipschitz-in- θ kernel.) $W_2(P_{\Omega|\theta_1}, P_{\Omega|\theta_2}) \leq L_{\text{lik}} \|\theta_1 - \theta_2\|$.

Then:

975 (a) (H2') holds on the goal-marginal post-update model law with $C_{T_2} \leq L_{\text{post}}^2 \max((1 +$
976 $2L_{\text{lik}}^2) C_\theta, 4/\rho_{\text{noise}})$.

977 (b) Track 1 yields $\mathbb{E} W_2^2(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq C_{T_2} I(G; M_{\tau+} | e_\tau, M_{\tau-})$.

978 (c) The cascade constant has the canonical Stuart-school form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$ across the Stuart-school
979 hypothesis space, with an absolute multiplicative overhead from the kernel-tensorization step. The
980 tight canonical constant $2L_{\text{post}}^2/\rho_{\text{noise}}$ is recovered on instances where the post-update parameter law
981 admits a direct sub-Gaussian read-off (conjugate-Gaussian and exponential-family-conjugate cases
982 — see Appendix C for the worked-through Class 1 instance).

983 *Proof.* By Lemma E.2 applied to (C0)+(C4a)+(C4b), the joint $P_\theta \otimes K \in T_2(C_J)$ on $\Theta \times \Omega$ with
984 $C_J \leq \max((1 + 2L_{\text{lik}}^2)C_\theta, 4/\rho_{\text{noise}})$. The projection $(\theta, \Omega) \mapsto \Omega$ is 1-Lipschitz, so by Lemma E.1,
985 $P_\Omega \in T_2(C_J)$. The map $\Omega \mapsto \mu^\Omega$ is L_{post} -Lipschitz by (C3), so by Lemma E.1 again, $P_{M_{\tau+}|e, M_{\tau-}} \in$
986 $T_2(L_{\text{post}}^2 C_J) = T_2(C_{T_2})$. (b) follows from Track 1's standard cascade (Appendix E.1). (c) is the
987 matching of canonical form; the tight constant on conjugate instances comes from a *direct* sub-
988 Gaussian read-off on the post-update parameter law (the structural Lipschitz-composition route is
989 blind to the Bayesian-update contraction). $\square$

990 Each Strand 2 paper — Stuart 2010^[7], Sprungk 2020^[8], Cvetković and Lie 2025^[9],
991 Dolera and Mainini 2023^[10], Garbuno-Iñigo et al. 2023^[11], Lie et al. 2018^[34],
992 Hosseini et al. 2025^[12] — establishes a form of (C3) under increasingly weak hypotheses.
993 Theorem E.3 is a structural slot they all fit into via the unbundled (C0)+(C3)+(C4a)+(C4b) hypothesis
994 set; anywhere the Stuart-school cascade applies, Track 1 specializes to a constant of the canonical
995 form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$. The result is *not* a sub-case statement — Stuart-school proves outputs (matching
996 lower bounds, multi-metric extensions, sub-Lipschitz refinements, contraction rates, randomized
997 forward models) outside Track 1's reach. The averaged-over-goal form of Track 1 also cannot recover
998 Stuart-school's pointwise pair-distance information on its own; the two literatures share machinery
999 and exchange a constant of the same form, not theorem closures. The averaged-vs-pointwise gap and
1000 the structural-vs-direct-sub-Gaussian gap on constant tightness are the two precise failure modes of
1001 the alternative reading “Track 1 contains the Stuart-school lineage as a strict sub-case.”

1002 E.3 Track 2 cascade — full derivation

1003 The umbrella theorem's Track 2 instantiation (under (H1) + (H4') + (PI) + (R) + (K)) follows in one
1004 step using the KL-to-Fisher-Rao expansion.

1005 **Lemma E.4** (KL-to-Fisher-Rao second-order expansion). *On the ambient L^2 -sphere of $\sqrt{p}$ equipped*
1006 *with the Fisher metric $\mathbf{I}$ (the maximal statistical manifold, of which every probability distribution is a*
1007 *point), for nearby P, Q : $\text{KL}(P \| Q) = \frac{1}{2} d_{FR}^2(P, Q) + O(d_{FR}^3)$ in the Amari-Nagaoka spherical-arc*
1008 *convention.*

1009 **Lemma E.5** (Uniform third-order remainder under (H4')). *Suppose the ambient statistical manifold*
1010 *along the goal-induced neighborhood admits a chart θ in which (i) the Amari-Chentsov tensor satisfies*
1011 *$|T_{ijk}(\theta^*) \delta^i \delta^j \delta^k| \leq \mathcal{T}_* \|\delta\|^3$ for all θ^* on the geodesic between θ and $\theta + \delta$, and (ii) the Fisher matrix*
1012 *is uniformly bounded below: $\mathbf{I}(\theta^*) \succeq \mathbf{I}_{\min} \cdot \text{Id}$ with $\mathbf{I}_{\min} > 0$, on the same neighborhood. Then for*
1013 *every pair (P_g, P_*) on the neighborhood with $\text{KL}(P_g \| P_*) \leq \delta_* \leq \mathbf{I}_{\min}^3/(8\mathcal{T}_*^2)$,*

$$1014 \quad d_{FR}^2(P_g, P_*) \leq 2 \text{KL}(P_g \| P_*) \cdot (1 + R_3(\delta_*)), \quad R_3(\delta_*) := \frac{2\mathcal{T}_* \sqrt{2\delta_* / \mathbf{I}_{\min}^3}}{1 - \mathcal{T}_* \sqrt{2\delta_* / \mathbf{I}_{\min}^3}} \xrightarrow{\delta_* \rightarrow 0} 0.$$

1015 *Proof. (of Lemma E.5.)* The Lagrange-form Taylor remainder for the KL expansion (Amari and
1016 Nagaoka¹⁹ §3.7 Theorem 3.1) gives, with hypothesis (i),

$$1017 \quad |\text{KL}(P_g \| P_*) - \frac{1}{2} \delta^\top \mathbf{I}(\theta^*) \delta| \leq \frac{1}{6} \mathcal{T}_* \|\delta\|^3,$$

1018 for some θ^* on the geodesic. Hypothesis (ii) gives $\delta^\top \mathbf{I}(\theta^*) \delta = d_{FR}^2(P_g, P_*)(1 + O(\|\delta\|))$ at leading
1019 order with the leading-order remainder also controlled by $\mathcal{T}_*/\mathbf{I}_{\min}^{3/2}$; folding both into a single inequality
1020 and using $d_{FR}^2 \geq \mathbf{I}_{\min} \|\delta\|^2$ (so $\|\delta\| \leq d_{FR}/\sqrt{\mathbf{I}_{\min}}$, $\|\delta\|^3 \leq d_{FR}^3/\mathbf{I}_{\min}^{3/2}$),

$$1021 \quad d_{FR}^2(P_g, P_*) \leq 2 \text{KL}(P_g \| P_*) + \frac{1}{3} \mathcal{T}_* d_{FR}^3/\mathbf{I}_{\min}^{3/2}.$$

Setting $u := d_{FR}(P_g, P_*) \geq 0$ and $K := \text{KL}(P_g \| P_*) \leq \delta_*$, this is $u^2 \leq 2K + (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2}$, i.e.,
 $u^2(1 - (\mathcal{T}_*/3)u/\mathbf{I}_{\min}^{3/2}) \leq 2K$. The polynomial $f(u) := u^2 - (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2} - 2K$ is increasing on
 $[0, 2\mathbf{I}_{\min}^{3/2}/\mathcal{T}_*]$ (where $f'(u) = u(2 - \mathcal{T}_*u/\mathbf{I}_{\min}^{3/2}) \geq 0$) and has a unique root $u^*(K)$ in that interval,
dominating u . Since $f(\sqrt{2K}) = -(\mathcal{T}_*/3)(2K)^{3/2}/\mathbf{I}_{\min}^{3/2} < 0$, the root satisfies $u^*(K) \geq \sqrt{2K}$; the
explicit upper bound from substituting $u \leq u^*(K)$ back into $u^2 \leq 2K + (\mathcal{T}_*/3)u^3/\mathbf{I}_{\min}^{3/2}$ and iterating
once gives $u^*(K)^2 \leq 2K(1 + R_3(K))$ with $R_3(K) = (2\mathcal{T}_*/3)\sqrt{2K}/\mathbf{I}_{\min}^{3/2}/(1 - (\mathcal{T}_*/3)\sqrt{2K}/\mathbf{I}_{\min}^{3/2})$
for $K \leq \mathbf{I}_{\min}^3/(8\mathcal{T}_*^2)$ (which keeps the denominator $\geq 2/3$). Hence $u^2 \leq 2K(1 + R_3(K)) \leq$
 $2K(1 + R_3(\delta_*))$, and $R_3(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$. $\square$

Proof. (Track 2 cascade.) Apply Lemma E.5 under (H1) + (H4') slice-wise at each $G = g$:

$$d_{FR}^2(P_{M_{\tau+}|e,M,G=g}, P_{M_{\tau+}|e,M}) \leq 2 \text{KL}(P_{M_{\tau+}|G=g} \| P_{M_{\tau+}}) (1 + R_3(\delta_*)),$$

with $R_3(\delta_*) \rightarrow 0$ as $\delta_* \rightarrow 0$. The (H4') regime supplies the slice-wise $\text{KL}_g \leq \delta_*$ premise; the
chart-existence premise (uniform Amari-Chentsov bound + minimum Fisher eigenvalue) is ambient-
Fisher-curvature regularity along the goal-induced neighborhood — implied by (H4') on smooth
statistical-manifold-sub-case (H1) families since both quantities depend continuously on the operating
point and the neighborhood is compactly generated.

Take expectation over G and substitute the chain-rule identity from Lemma 5.1:

$$\mathbb{E}[d_{FR}^2(P_{M_{\tau+}|G}, P_{M_{\tau+}})] \leq 2 \cdot I(G; M_{\tau+} | e_{\tau}, M_{\tau-}) \cdot (1 + o(1)).$$

Jensen on $\sqrt{x}$ delivers the distance form

$$\mathbb{E} d_{FR}(P_{M_{\tau+}|G}, P_{M_{\tau+}}) \leq \sqrt{2} \cdot \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})} \cdot (1 + o(1)). \quad \square$$

The constant $\sqrt{2}$ emerges from the second-order coefficient $\frac{1}{2}$ in the KL-to-Fisher expansion (Amari
and Nagaoka¹⁹ §3.7 Theorem 3.1) — exactly what (K) crystallizes as a normalization. Outside (H4'),
the locally-tight bound's $(1 + o(1))$ remainder is no longer uniformly sharp; companion bounds in
Appendix D cover the moderate-to-large I_M regime under strictly weaker hypotheses (Track 2 global,
Hellinger backstop).

E.4 No-go on Euclidean chart norms — full proof

Theorem 4.4's proof uses a chart-rescaling lemma plus a one-line consequence.

Lemma E.6 (Chart-rescaling sensitivity of Euclidean displacement). *Let $\mathcal{M}$ be a statistical manifold
and $\phi : U \rightarrow \mathbb{R}^d$ a chart on an open set $U \subseteq \mathcal{M}$. For any $a > 0$, write $\phi_a := a\phi$ for the
rescaled chart. For every pair of probability laws μ, ν on U (in particular, the goal-conditional
and goal-marginal post-update laws of (3)): (a) The chart-Euclidean Wasserstein distance scales
linearly: $W_2^{\phi_a}(\mu, \nu) = a \cdot W_2^{\phi}(\mu, \nu)$. (b) Conditional KL, conditional mutual information, Fisher-Rao
spherical-arc distance, and Hellinger distance are all chart-invariant: they take the same value under
 ϕ and ϕ_a .*

Proof. (a) The pushforward under the linear map $x \mapsto ax$ rescales Euclidean distances by a , hence
rescales any optimal coupling's transport cost. (b) KL depends only on the underlying densities
and dominating measure (Radon-Nikodym derivatives are reparameterization-invariant). The Fisher
metric tensor transforms covariantly under chart change, so the integrated geodesic distance is invariant.
Hellinger as an f -divergence is reparameterization-invariant. $\square$

Proof. Proof of Theorem 4.4. Suppose for contradiction that a coordinate-independent $C_0 < \infty$
exists such that for every chart ϕ on $\mathcal{M}$ and every architecture satisfying (H1)+(H2') with non-zero
post-update displacement,

$$\|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} \leq C_0 \sqrt{I(G; M_{\tau+} | e_{\tau}, M_{\tau-})}.$$

For any architecture realizing $W_2^{\phi}(P_{M_{\tau+}|G,e,M_{\tau-}}, P_{M_{\tau+}|e,M_{\tau-}}) > 0$ and finite positive transferred
information, applying the supposed bound under the rescaled chart ϕ_a gives, by Lemma E.6,

$$a \cdot \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi} = \|\Delta M_{\text{bias}}\|_{\text{Eucl}, \phi_\alpha} \leq C_0 \sqrt{I},$$

with the right-hand side unchanged across charts. Taking $a \rightarrow \infty$ contradicts the fixed $C_0 \sqrt{I}$. The same conclusion holds for the architectural-corollary form $C \sqrt{\kappa_{\text{processing}} \cdot I(G; \Omega \mid e, M)}$ under (H_κ) : both sides inherit Lemma E.6’s chart-(non)invariance, and the contradiction goes through verbatim. $\square$

The proof avoids type-shifts between Dirac points and pushforward laws, never invokes $(H2')$ -violating two-point constructions, and does not assert any specific value of I — it only uses that some architecture realizes finite positive W_2^ϕ and finite positive I , automatic on any non-trivial goal-coupled architecture (e.g., the conjugate-Gaussian setup with any $\sigma, \tau > 0$ and a non-degenerate continuous goal distribution). The Gaussian scale family with two charts (Chart A: σ ; Chart B: $\log \sigma$) is the canonical illustration: same intrinsic geometry, Chart-A Euclidean displacement grows linearly while Chart-B stays bounded as the operating scale grows.

E.5 Čencov uniqueness and sharpness — full proof

Theorem 4.5 has two parts.

Proof. Proof of (a) — Uniqueness of the metric. (PI) + (R) invokes Čencov’s uniqueness theorem^[18] (modern treatment Ay et al.³⁵, Theorem 5.1) on the *categorical metric assignment* of (PI): the assignment must give the Fisher information metric $\mathbf{I}$ on every standard parametric statistical model, up to a single global positive scalar $c > 0$. The induced metric on the model family at hand satisfies $d_{\mathcal{M}} = \sqrt{c} d_{FR}$ where d_{FR} is at the Amari-Nagaoka normalization $\mathbf{I}(\theta) = -\mathbb{E}[\nabla^2 \log p_\theta]$. The KL-to-Fisher second-order expansion (Amari and Nagaoka¹⁹ §3.7 Theorem 3.1) gives $\text{KL}(P_\theta \| P_{\theta+\delta}) = \frac{1}{2} d_{FR}^2 + O(d_{FR}^3)$, so $\frac{1}{2} d_{\mathcal{M}}^2 = (c/2) d_{FR}^2 + O(d_{FR}^3)$. Constraint (K) requires $\frac{1}{2} d_{\mathcal{M}}^2$ agree with KL at second order, forcing $c = 1$ and $d_{\mathcal{M}} = d_{FR}$. $\square$

Proof. Proof of (b) — Sharpness of the constant. Use the symmetric two-point witness on the conjugate-Gaussian Class 1 (Separated) family. Take $G \in \{-a, +a\}$ uniform with $\beta(g) = g$, prior $\theta \sim \mathcal{N}(0, \tau^2)$, observation $\Omega \mid G = g \sim \mathcal{N}(g, \sigma^2)$, and posterior $M_{\tau+} = L_{\text{post}} \Omega$. Then $P_{M_{\tau+} | G=g} = \mathcal{N}(L_{\text{post}} g, L_{\text{post}}^2 \sigma^2)$ and the goal-marginal is the equal-weight mixture of these two Gaussians (centered at $\pm L_{\text{post}} a$). For fixed σ and $a/\sigma \rightarrow 0$ the mixture’s variance excess is $L_{\text{post}}^2 a^2 = O((a/\sigma)^2) \cdot \sigma^2$ — second-order in a/σ — while the mean-shift is first order in a/σ , so the mixture is approximately Gaussian at $\mathcal{N}(0, L_{\text{post}}^2 \sigma^2)$ to leading order. The Fisher-Rao distance between two equal-variance univariate Gaussians is $|\mu_1 - \mu_2|/\sigma_{\text{shared}}$, giving $d_{FR}(P_{M_{\tau+} | G=\pm a}, P_{M_{\tau+}}) = a/\sigma$ at leading order, *exactly equal across both goal slices by symmetry*. Hence $\text{Var}_G(d_{FR}) = O((a/\sigma)^4)$ and Jensen is tight in the small-SNR limit. The transferred information for binary-Gaussian channels at small SNR is $I_M = (a/\sigma)^2/2 + O((a/\sigma)^4)$. Therefore $\mathbb{E} d_{FR}/\sqrt{I_M} = (a/\sigma)/\sqrt{(a/\sigma)^2/2} \cdot (1 + o(1)) = \sqrt{2} (1 + o(1))$ as $a/\sigma \rightarrow 0$, and equivalently $\mathbb{E} d_{FR}^2/I_M \rightarrow 2$. The sharpness holds at the first-moment level — Jensen is asymptotically tight by the symmetry-induced near-constancy of d_{FR} across slices — so any candidate constant $C < \sqrt{2}$ fails on this family. Combined with (a), $C = \sqrt{2}$ is the unique sharp upper-bound constant. $\square$

The upshot: once we adopt (PI) at full Markov-morphism strength, plus Riemannian structure, plus KL as the cascade’s information coordinate, the *only* such-invariant choice of norm on $\mathcal{M}$ is Fisher-Rao, and the *unique sharp* upper-bound constant is $\sqrt{2}$. Alternative chart-independent commitments — TV with Pinsker, Hellinger-as-divergence, χ^2 , Rényi — give weaker bounds: TV-Pinsker is non-tight (the constant is universal but the inequality is loose except at coincident measures); Hellinger-as-divergence and χ^2 -as-divergence violate (R); switching divergences on the RHS violates (K). The global Theorem D.1’s $C = 2$ is also sharp in its own (PI)-only regime — approached arbitrarily closely as $N \rightarrow \infty$ on the symmetric N -point family of Section D.1.1.

E.6 Coupled-class autoregressive connectivity — full statement and proof

Lemma E.7 (Coupled-class autoregressive connectivity). *Let $\mathcal{G}_\theta$ be the directed computational graph of an autoregressive sequence model with parameters θ . Suppose at every layer $\ell \geq 1$ each*

per-position update factors as $h_\ell^{(j)} = h_{\ell-1}^{(j)} + F_\ell^{(j)}(\{h_{\ell-1}^{(i)}\}_{i \leq j}; \theta) + G_\ell(h_{\ell-1}^{(j)}; \theta)$, with $F_\ell^{(j)}$ a causally-aggregated cross-position mixer and G_ℓ position-wise. If goal positions i_G and evidence positions i_E satisfy $i_G \cap i_E = \emptyset$, then for every layer $\ell \geq 1$ and position $j \geq \max(i_G \cup i_E)$, $h_\ell^{(j)}$ has a directed-graph path back to every $i_G \leq j$ whenever the per-source non-degeneracy condition $(\star)$ holds: for each $i \leq j$, $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)}$ is generically non-zero on the operating set.

Corollary E.8 (Named-architecture instantiations). *Per-source non-degeneracy is satisfied generically by: (a) plain decoder-only transformer attention (non-degenerate softmax attention weights, non-singular value projection); (b) linear-attention transformers^[36] with non-vanishing feature map (e.g., $\phi = \text{elu} + 1$, $\phi = \exp$); (c) selective state-space models / Mamba^[37] (non-singular discretized $\bar{A} = \exp(\Delta A)$ with $A < 0$ diagonal, non-zero B and C); (d) RWKV time-mixing^[38] (positive time-decay); (e) RetNet^[39] (decay $\gamma \in (0, 1)$); (f) long-convolution / Hyena architectures^[40] (full-support causal kernel).*

Each is structurally Class 3 (Coupled) under standard parameterization.

The proof is a one-induction over layer depth, structurally identical across the architectures of Corollary E.8 — the per-source non-degeneracy condition is the architecture-specific input, the directed-graph-path conclusion is the architecture-agnostic output. Robust to RMSNorm^[41] / Group-Norm^[42] / FlashAttention^[43] (position-wise / mathematically-equivalent), causal masking (preserves $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i_G)} \neq 0$ for $i_G \leq j$), and sliding-window or sparse attention (preserves connectivity by composition across layers when the window pattern admits a multi-hop path). Architectures with explicit goal-to-output blocking — token-dropping at intermediate layers, hard-routing MoE-attention with goal-conditional partitioning, or strictly goal-blind retrieval components — fall outside the lemma’s scope and are appropriately Class 2 (Partial).

Proof. Proof of Lemma E.7. By induction on layer depth, in the unified causally-aggregated-mixer form.

Layer 0 (embedding). $h_0^{(i)}$ depends only on X_i . Goal embeddings $h_0^{(i_G)}$ are functions only of the goal tokens at those positions; same for evidence positions i_E . No cross-position dependence yet.

Layer $\ell \geq 1$. The per-position update $h_\ell^{(j)} = h_{\ell-1}^{(j)} + F_\ell^{(j)}(\{h_{\ell-1}^{(i)}\}_{i \leq j}; \theta) + G_\ell(h_{\ell-1}^{(j)}; \theta)$ induces a directed position-graph with two edge types: (i) *same-position* residual edge $h_{\ell-1}^{(j)} \rightarrow h_\ell^{(j)}$ always present; (ii) *cross-position* mixer edges $h_{\ell-1}^{(i)} \rightarrow h_\ell^{(j)}$ present whenever $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} \neq 0$, i.e., whenever $(\star)$ holds at (j, i) .

Inductive claim. For every $\ell \geq 0$ and every input position $i \leq j$ along which $(\star)$ holds at some intermediate layer, $h_\ell^{(j)}$ has a directed-graph path back to input position i .

Inductive step. The same-position residual chain gives $i = j$. For $i \neq j$ with $i \leq j$, $(\star)$ at any single layer $\ell' \leq \ell$ along the path gives a cross-position edge $h_{\ell'-1}^{(i)} \rightarrow h_{\ell'}^{(j)}$, which composes with the residual chain on either side to yield a directed-graph path from input position i to $h_\ell^{(j)}$.

Coupled-class conclusion. For every goal position $i_G \leq j$, $h_\ell^{(j)}$ has a path back to position i_G whenever $(\star)$ holds at (j, i_G) at some layer. So G (encoded at positions i_G) is causally upstream of every quantity contributing to a post-update model state read off from position $j \geq \max(i_G \cup i_E)$. Class 3 (Coupled) by construction. $\square$

Per-architecture verifications of $(\star)$ Each architecture in Corollary E.8 satisfies $(\star)$ generically; the verifications are short and architecture-specific.

(a) Self-attention. $F_\ell^{(j)} = \sum_i \alpha_{ji}^{(\ell)}(h_{\ell-1}) V_\ell h_{\ell-1}^{(i)}$ with softmax weights $\alpha_{ji} > 0$. Per-source partial $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} = \alpha_{ji}^{(\ell)} V_\ell + \text{softmax-derivative cross-terms}$. Strictly positive softmax + generically non-trivial value-projection action ($V_\ell h_{\ell-1}^{(i)} \neq 0$ on the operating set, or equivalently V_ℓ has no left-null subspace containing the operating-set hidden states) $\Rightarrow (\star)$.

1159 (b) *Linear attention*^[36]. $F_\ell^{(j)}$ is a kernel-weighted sum with per-source factor
 1160 $\phi(Kh^{(i)})^\top \phi(Qh^{(j)})Vh^{(i)}$. $(\star)$ holds when ϕ is non-vanishing on the operating set: standard choices
 1161 $\phi = \text{elu} + 1$, $\phi = \exp$ are strictly positive.

1162 (c) *Selective state-space models / Mamba*. Unrolling the recurrence: $y_\ell^{(j)} =$
 1163 $C_\ell^{(j)} \sum_{i \leq j} (\prod_{k=i+1}^j \bar{A}_\ell^{(k)}) \bar{B}_\ell^{(i)} x_\ell^{(i)}$. Per-source partial leading factor $C_\ell^{(j)} \prod \bar{A} \cdot \bar{B}_\ell^{(i)}$. Under standard
 1164 parameterization $\bar{A} = \exp(\Delta A)$ with $A < 0$ diagonal (HiPPO-style) and $\Delta = \text{softplus}(\cdot) > 0$,
 1165 eigenvalues lie in $(0, 1)$ — non-singular for every k . $\bar{B}$ and C generically non-zero. Selectivity
 1166 is *gain modulation* in a one-parameter family, not a path-cutting routing decision; quantitative
 1167 attenuation lives in $\kappa_{\text{processing}}$.

1168 (d) *RWKV*. Time-mixing $\text{wkv}_t = \frac{\sum_{i \leq t} e^{-(t-1-i)w+k_i} v_i + e^{u+k_t} w_t}{\sum_{i < t} e^{-(t-1-i)w+k_i} + e^{u+k_t}}$ has strictly-positive per-source weight
 1169 $e^{-(t-1-i)w+k_i}$. $(\star)$ holds.

1170 (e) *RetNet*. $S_n = \gamma S_{n-1} + k_n^\top v_n$, $y_n = q_n S_n$ with $\gamma \in (0, 1)$ unrolls to $y_n = q_n \sum_{i \leq n} \gamma^{n-i} k_i^\top v_i$.
 1171 Per-source partial $q_n \gamma^{n-i} k_i^\top$ generically non-zero.

1172 (f) *Long-convolution / Hyena*. $y_j = \sum_{i \leq j} h_{j-i} x_i$. Per-source partial is the kernel value h_{j-i} ,
 1173 non-zero under full causal support.

1174 *Robustness to common variants*. RMSNorm^[41], GroupNorm^[42], and FlashAttention^[43] preserve the
 1175 position-graph (position-wise normalization or mathematically-equivalent attention). Causal masking
 1176 preserves $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i_G)} \neq 0$ for $i_G \leq j$. Sliding-window or sparse attention preserves connectivity
 1177 by multi-hop composition when the window pattern admits a path.

1178 *Class 2 boundary*. Hard-routing MoE-attention with goal-conditional partitioning and token-dropping
 1179 at intermediate layers fall outside $(\star)$ structurally — the goal-to-output graph path is severed by
 1180 hypothesis. Vanishing-feature-map linear attention ($\phi(x) = \text{ReLU}(x)$ on negative inputs) and SSM
 1181 near-state-collapse (Δ large but finite) are *quantitative* boundaries: the directed-graph path remains,
 1182 $(\star)$'s genericity holds, and structural Class 3 is intact, but $\kappa_{\text{processing}}$ is small. *Exact* state-collapse
 1183 ($\bar{A} \rightarrow 0$, $\Delta \rightarrow \infty$) is the limiting case where the cross-position partial $\partial F_\ell^{(j)} / \partial h_{\ell-1}^{(i)} \rightarrow 0$ exits $(\star)$'s
 1184 genericity — beyond the lemma's quantitative certification but reachable only in the limit, not for any
 1185 finite-magnitude Δ . Strictly goal-blind retrieval components yield system-level Class 2.

1186 Lemma E.7 establishes a *downstream-output graph-reachability* claim: at every layer and every
 1187 position causally downstream of the goal and evidence under standard causal masking, the activation
 1188 has a directed-graph path back to at least one goal position. *Three caveats*:

- 1189 • *Graph reachability is not the same as causal-influence magnitude*. A directed-graph path
 1190 with near-zero attention weights produces a formal edge with negligible quantitative influence
 1191 on $h_\ell^{(j)}$. The connectivity claim is therefore *structural* — about the existence of a path,
 1192 not the magnitude of effect — and translates into Class 3 (Coupled) classification under
 1193 the Markov-blanket conditional-independence reading. Bounding $\kappa_{\text{processing}}$ requires the
 1194 architectural-bandwidth conditions of Appendix B.3 (information-bottleneck on the goal
 1195 channel; bounded direct-channel capacity uniformly over operating distributions).
- 1196 • *Causal masking restricts the lemma's scope to downstream positions*. For $j < \min(i_G)$ —
 1197 positions strictly upstream of any goal token — there is no causal path from goal to $h_\ell^{(j)}$
 1198 under causal attention masks; those positions are Class 1 (Separated) in their own right.
- 1199 • *Sparse and sliding-window attention require a multi-hop reachability assumption*. The
 1200 connectivity proof composes single-layer attention edges across layers; for sliding-window
 1201 or sparse-attention architectures, multi-hop connectivity from i_G to j requires at least one
 1202 composition of single-layer paths through the window/sparsity pattern. Architectures whose
 1203 sparsity pattern excludes goal-to-output communication entirely fall outside the lemma's
 1204 hypothesis and are Class 2 (Partial).

1205 The bias-bound *interpretation* — bounding the displacement of an implicit posterior under goal-
 1206 conditional re-prompting — additionally invokes the in-context-learning correspondence^[13–16] be-
 1207 tween the next-token distribution and an implicit posterior. The structural Coupled-class claim of
 1208 Lemma E.7 is robust without it; the bias-bound *application* is conditional on the in-context-learning

1209 correspondence, which has its own scope (in-distribution prompts: tight; OOD/jailbreak: degraded to
1210 a bound on output-distribution displacement rather than belief-state displacement).

1211 **F Extended related work**

1212 §2 Related Work names the two main lineages this paper composes (LLM hallucination theory;
1213 Bayesian inverse-problems posterior stability). This appendix covers the additional adjacent strands —
1214 Fisher-Rao Bayesian sensitivity, Owhadi-Scovel-Sullivan brittleness, mean-field variational inference
1215 stability, and the Markov-blanket apparatus from active inference — that sit close enough to require
1216 explicit cite-and-distinguish but do not resolve the goal-coupling-displacement gap.

1217 **F.1 Strand 1: LLM hallucination theory**

1218 The recent statistical-inevitability and capacity-impossibility lineage targets the *frequency* of hallucina-
1219 tion on architecture-agnostic grounds. Kalai and Vempala 2024^[2] establish that calibrated language
1220 models must hallucinate at a rate close to the fraction of singleton facts in the training data — a
1221 statistical lower bound that holds independently of the transformer architecture. Kalai et al. 2026^[3]
1222 extend this to argue that hallucinations originate as binary-classification errors compounded by
1223 training-and-evaluation incentives that reward guessing under uncertainty. Karbasi et al. 2025^[4]
1224 reduce hallucination *detection* to language identification in the Gold-Angluin framework, showing
1225 that detection is impossible from positive examples alone but feasible with expert-labeled negatives.
1226 Wu et al. 2025^[5] establish a VC-dimension-based no-free-lunch result for non-hallucinating learning.
1227 Suzuki et al. 2025^[6] give a probabilistic positive result — hallucinations can be made statistically neg-
1228 ligible — counterbalancing the computability-theoretic inevitability arguments. Guo and Li 2026^[44]
1229 frame hallucination as a rate-distortion theorem for membership testing under capacity constraints.
1230 Liu et al. 2025^[45] formalize hallucinations as estimation failures with high-probability lower bounds
1231 on the hallucinate rate. Chlon et al. 2025^[46] frame transformers as Bayesian-in-expectation-not-
1232 realization with $O(\log n)$ permutation dispersion. Zeng et al. 2026^[47] decompose hallucination risk
1233 into data-driven and reasoning-driven components.

1234 Our distance from this strand: we bound a *displacement*, not a frequency, under named *geometric*
1235 assumptions, and we engage the *architecture’s coupling structure* directly via $\kappa_{\text{processing}}$. The bound is
1236 not a substitute for the frequency bounds; it is an orthogonal axis. Among the recent papers, Zeng
1237 et al.’s decomposition is closest in shape to our $\kappa \times I$ factorization but differs on structural axis
1238 (training-time vs inference-time, vs our architectural-vs-informational).

1239 **F.2 Strand 2: Bayesian inverse-problems posterior stability**

1240 The transport-inequality and Lipschitz-posterior machinery driving Track 1 is mature. Stuart 2010^[7]
1241 establishes well-posedness of Bayesian inverse problems on function spaces, with Lipschitz continuity
1242 of the posterior in the data. Sprungk 2020^[8] extends this to local Lipschitz stability with respect
1243 to perturbations of the prior and the log-likelihood, in TV / Hellinger / Wasserstein / KL distances.
1244 Cvetković and Lie 2025^[9] match the upper bounds with lower bounds, emphasizing that posterior
1245 sensitivity *increases* as the posterior concentrates. Dolera and Mainini 2023^[10] obtain Lipschitz-
1246 continuity of probability kernels in the optimal-transport framework, with explicit constants in
1247 terms of Fisher-information functionals and weighted Poincaré constants — the closest-in-spirit
1248 prior result to Track 1’s form. Garbuno-Iñigo et al. 2023^[11] develop integral-probability-metric
1249 perturbation analysis under log-Sobolev-style functional inequalities. Latz 2020^[48] provides well-
1250 posedness under weaker continuity. Lie et al. 2018^[34] give Hellinger-distance bounds for randomized
1251 forward models. Hosseini et al. 2025^[12] develop conditional optimal transport on function spaces
1252 for amortized Bayesian inference, with explicit regularity estimates on conditioning maps from prior
1253 to posterior — the closest single-paper match for Track 1’s cascade. Dolera et al. 2024^[49] derive
1254 Wasserstein-based posterior contraction rates. Del Moral 2025^[50] gives Sinkhorn / Schrödinger
1255 bridge contraction via log-Sobolev — most recent transport-inequality-plus-LSI work, adjacent to
1256 Track 1’s machinery.

1257 Each of these results establishes a form of Lipschitz-in-data continuity of the posterior under in-
1258 creasingly weak hypotheses. Theorem E.3 shows that any such Lipschitz-posterior result, combined
1259 with T_2 concentration on the goal-induced parameter measure and slice-wise sub-Gaussianity of the

conditional data law, is a sufficient verification path for Track 1’s (H2’) on the post-update model law in any Class 1 (Separated) architecture, with cascade constant of the canonical Stuart-school form $\propto L_{\text{post}}^2/\rho_{\text{noise}}$ — tight at $2L_{\text{post}}^2/\rho_{\text{noise}}$ on conjugate-Gaussian and exponential-family-conjugate instances, with absolute multiplicative overhead from the structural kernel-tensorization step in general. Hosseini-Hsu-Taghvaei is the closest single-paper match for the amortized-inference framing; the structural-slot match holds for the entire strand.

Our distance from this strand: we apply this shared machinery to *belief-goal-coupled* inference under an architectural classification. None of these papers address goal-conditional bias under explicit architectural-coupling factorization. The relationship is shared-machinery, not theorem-containment — Strand 2 proves outputs (matching lower bounds, multi-metric extensions, sub-Lipschitz refinements, contraction rates, randomized forward models) outside Track 1’s averaged-over-goal reach.

1271 F.3 Strand 3: Fisher-Rao geometry for Bayesian sensitivity

Kurtek and Bharath 2015^[51] provide a Fisher-Rao framework for Bayesian sensitivity analysis under ε -contamination classes — the closest existing Fisher-Rao deployment to ours. Same geometric machinery; different target (sensitivity vs universal-constant bias bound). Lebanon 2004^[52] extends Čencov-Campbell uniqueness to conditional information geometry — load-bearing precursor for the conditional-distribution setting in which our bound lives. Čencov 1982^[18] is the original uniqueness result for the Fisher metric under invariance under sufficient statistics; Amari and Nagaoka 2000^[19] give the modern treatment of the KL-to-Fisher second-order expansion. McCulloch 1989^[53] and Ruggeri and Wasserman 1993^[54] develop earlier Fisher-information-flavored sensitivity analyses; both are foundational but produce neither a universal constant nor a goal-coupling bound.

1281 F.4 Strand 4: Bayesian brittleness (handle carefully)

Owhadi et al., Owhadi et al. 2015, 2015^[24,25] establish that Bayesian inference is brittle under finite information in continuous settings: arbitrarily close models on the same data can yield arbitrarily different posterior conclusions, and any prior + model can be perturbed slightly to achieve any desired posterior. Owhadi and Scovel 2017^[55] develops the qualitative-robustness companion analysis.

A reader might worry that the brittleness results contradict our Track 1 stability bound. They do not. Owhadi-Scovel-Sullivan’s no-go is for *arbitrary stability under finite information* in fixed metrics (TV, generalized moments) — they construct adversarial perturbations *outside* the well-posed regime. Our Theorem 4.1 (Track 1 instantiation) carries explicit (H1) + (H2’) hypotheses that exclude their brittleness regime: (H2’)’s post-update concentration is precisely the regularity that brittleness requires *failing*. The two no-gos constrain different things; our Theorem 4.4 is for *universal-constant absent the (PI) + (R) + (K) triple*, which is a third, distinct statement. The honest reading: three different no-go results, three different scope conditions, none implying or contradicting the others.

1294 F.5 Strand 5: Adjacent recent work

Sheng et al. 2025^[56] establish stability of mean-field variational inference under strongly log-concave targets — adjacent to Track 1’s transport-inequality machinery. Cattiaux and Guillin 2022^[57] develop functional inequalities (Poincaré, Cheeger, log-Sobolev) for perturbed measures with explicit constants — background machinery for Hypothesis 4.2. Ley et al. 2017^[58] give Wasserstein-distance bounds for nested densities via the Stein method. Cvetković et al. 2024^[59] address observation-operator choice for misspecified inverse problems. Zahm et al. 2022^[60] provide certified dimension reduction for nonlinear Bayesian inverse problems. Kleijn and van der Vaart 2006^[61] study misspecified Bayesian inference asymptotics; Shalizi 2009^[62] studies dynamics of Bayesian updating under dependent data and misspecified models. Seo et al. 2026^[63] derive W_2 -stability bounds for tangent-linearized Gaussian inference on smooth manifolds — adjacent for the manifold-Bayesian-inference framing.

In LLM-stability-flavored work: Xu 2025^[64] (*Policy Cliff*) analyzes RLHF stability for LLMs/LRMs through reward-policy maps; Biau and Boyer 2026^[65] develop statistical proxies for retrieval-augmented generation; Hyeon et al. 2026^[66] address action-sufficient goal representations in RL. None of these target the goal-coupling bias quantity (3) under an architectural-coupling factorization.

1309 F.6 Strand 6: Active inference and Markov-blanket apparatus

1310 The Markov-blanket apparatus from active inference^[67–69] supplies the conditional-independence
1311 machinery our architectural-separation classification invokes. We adopt the Pearl-blanket reading^[17]
1312 — the technical conditional-independence statement — and explicitly do not adopt the Friston-blanket
1313 metaphysical reading. The Goal/Update Coupling Class partition (Class 1 Separated / Class 2 Partial
1314 / Class 3 Coupled), with the explicit Coupled-class scope-exit, is the structural extension that the
1315 Pearl-blanket form invites but does not by itself produce.

1316 F.7 Relation to existing decompositions

1317 Zeng et al. 2026^[47]’s HalluGuard framework decomposes hallucination risk into *data-driven* and
1318 *reasoning-driven* components, linked respectively to training-time mismatches and inference-time
1319 instabilities. This is the closest-shape decomposition to our $\kappa_{\text{processing}} \times I$ factorization. The structural
1320 axes differ: HalluGuard’s axis is *temporal* (training-time vs inference-time errors); ours is *architectural-*
1321 *vs-informational* (where the goal can flow vs how much it could resolve). The two decompositions live
1322 at different layers and do not reduce to each other. A model can have low data-driven hallucination
1323 (clean training) and still exhibit large goal-coupling displacement (Coupled-class architecture in a
1324 high-ambiguity regime), and vice versa.

1325 Chlon et al. 2025^[46] frame transformers as *Bayesian in expectation but not in realization*, with
1326 permutation-induced order effects scaling as $O(\log n)$. This is structurally distinct from our framing
1327 — we treat the architecture as performing a single goal-conditioned belief update on a fixed input,
1328 not as averaging over orderings — but the two are compatible: their Quantified Martingale Violation
1329 bounds the deviation from Bayesian-in-realization behavior, and our bound captures one specific
1330 channel (goal-coupling) of how non-Bayesian-in-realization a coupled architecture can be.

1331 F.8 Failed routes — pointer

1332 Two derivation routes failed at structural levels and are documented in Appendix A: (F1) Cramér-Rao
1333 inversion fails because Cramér-Rao bounds estimator variance from below, and inverting it for an upper
1334 bound on side-information-induced displacement requires a fixed-estimator-class assumption not
1335 available in coupled-architecture settings. (F2) Rate-distortion inversion fails because rate-distortion
1336 theory describes the minimum bits-per-symbol required to *represent* a source within distortion D ,
1337 not the maximum *displacement induced* by injecting side-information into an update. The pattern:
1338 information-theoretic *lower-bound* machinery for estimator or encoder performance does not invert to
1339 upper bounds on side-information-induced displacement.

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Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 state the four contributions: (i) the architectural Goal/Update Coupling Class partition, (ii) the umbrella upper bound on goal-conditional displacement of the post-update model state with two named tracks supplying the constant, (iii) the no-go theorem on Euclidean chart norms with the locally-tight $\sqrt{2}$ and globally-valid $C = 2$ Fisher-Rao constants forced under (PI)+(R)+(K) and (PI) respectively, and (iv) the architectural-corollary factorization $\kappa_{\text{processing}} \times I(G; \Omega \mid e, M)$ under (H_κ) . Each is developed formally in Sections 3–5 and proved in full in Appendix E.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
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